

Real-Life Problems

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Differentiation (ref d5)

This document contains all 70 *Real-Life Problems* questions (kinematics, optimisation, and rates of change), each with a fully worked solution and a TikZ graph of the model: displacement/area/volume/cost/etc. plotted against its variable, with the optimum or rest point marked, and a dashed **tangent** drawn for the instantaneous-rate questions (velocity, marginal cost/revenue, flow rate).

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Real-Life Problems 01 (d5-001)

Difficulty: Foundation / Marks: 3

A particle moves along a straight line. Its displacement s metres from a fixed point after t seconds is given by $s = 4t^2 - 3t$. (a) Find the velocity v as a function of t . (b) Find the velocity when $t = 5$ seconds.

Worked solution.

Step 1: (a) Velocity is the rate of change of displacement: $v = \frac{ds}{dt}$.

$$v = \frac{ds}{dt} = 8t - 3$$

Differentiate $s = 4t^2 - 3t$ with respect to t .

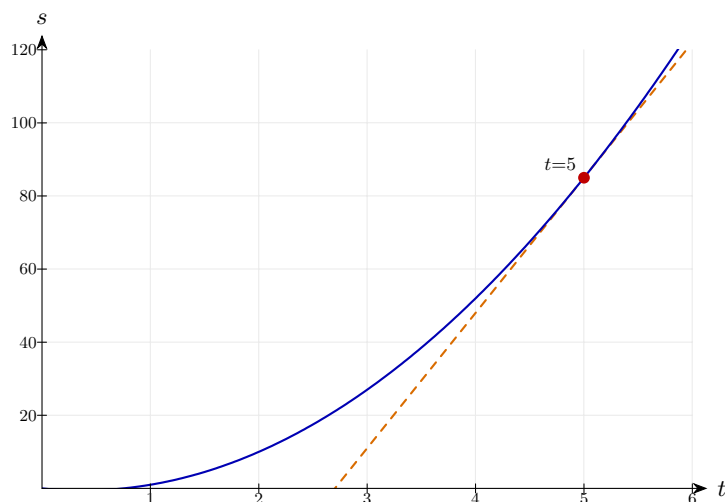
Step 2: (b) Substitute $t = 5$.

$$v = 8(5) - 3 = 40 - 3 = 37 \text{ ms}^{-1}$$

Replace t with 5 in the velocity expression.

Final answer: (a) $v = 8t - 3$
 (b) $v = 37 \text{ ms}^{-1}$

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 02 (d5-002)

Difficulty: Foundation / Marks: 4

A particle's displacement is $s = t^3 - 6t^2 + 5$ metres after t seconds. (a) Find the velocity function. (b) Find the acceleration function. (c) Find the acceleration when $t = 4$ seconds.

Worked solution.

Step 1: (a) Differentiate s to get velocity.

$$v = \frac{ds}{dt} = 3t^2 - 12t$$

Power rule applied term by term.

Step 2: (b) Differentiate v to get acceleration.

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2} = 6t - 12$$

Acceleration is the rate of change of velocity.

Step 3: (c) Substitute $t = 4$.

$$a = 6(4) - 12 = 24 - 12 = 12 \text{ ms}^{-2}$$

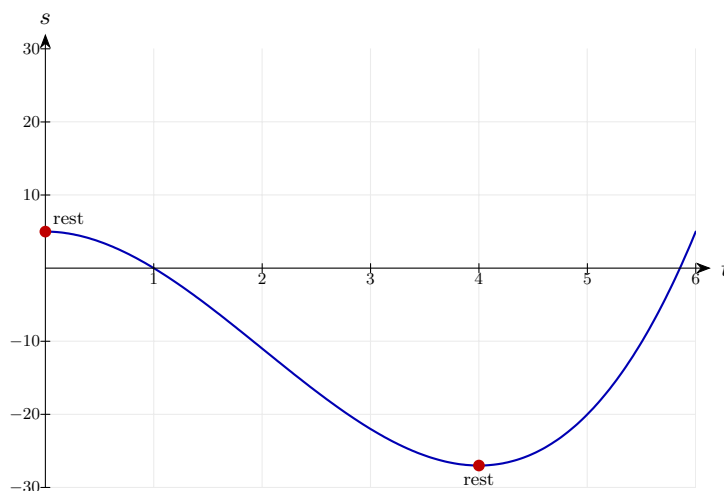
Replace t with 4.

Final answer: (a) $v = 3t^2 - 12t$

(b) $a = 6t - 12$

(c) $a = 12 \text{ ms}^{-2}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 03 (d5-003)

Difficulty: Foundation / Marks: 4

A stone is thrown upwards. Its height h metres above the ground after t seconds is $h = 20t - 5t^2$. (a) Find the velocity of the stone after t seconds. (b) Find the time at which the stone is momentarily stationary (at its maximum height). (c) Find the maximum height reached.

Worked solution.

Step 1: (a) Differentiate h with respect to t .

$$v = \frac{dh}{dt} = 20 - 10t$$

Velocity is the derivative of displacement (height).

Step 2: (b) At maximum height the velocity is zero.

$$20 - 10t = 0 \implies t = 2 \text{ s}$$

Set $v = 0$ and solve for t .

Step 3: (c) Substitute $t = 2$ into h .

$$h = 20(2) - 5(4) = 40 - 20 = 20 \text{ m}$$

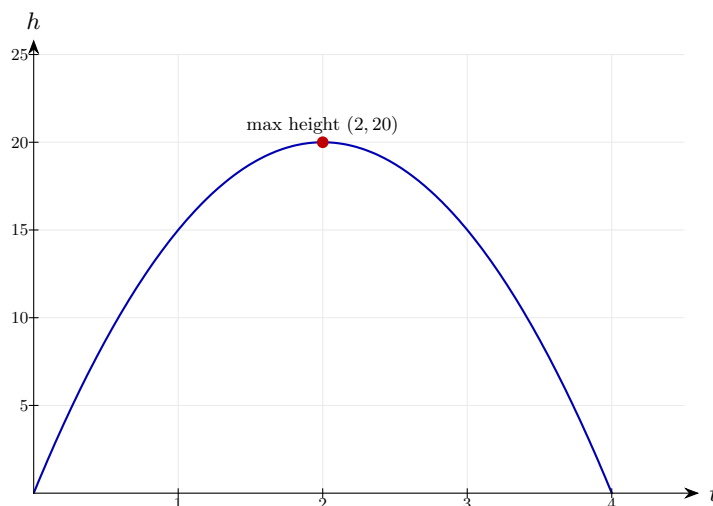
The maximum height occurs at $t = 2$.

Final answer: (a) $v = 20 - 10t$

(b) $t = 2 \text{ s}$

(c) Maximum height = 20 m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 04 (d5-004)

Difficulty: Foundation / Marks: 4

A particle moves along a straight line. Its displacement s metres after t seconds is $s = t^3 - 9t^2 + 24t$ for $t \geq 0$. (a) Find the velocity function $v(t)$. (b) Find the values of t at which the particle is momentarily at rest.

Worked solution.

Step 1: (a) Differentiate.

$$v = \frac{ds}{dt} = 3t^2 - 18t + 24$$

Power rule.

Step 2: (b) Set $v = 0$.

$$3t^2 - 18t + 24 = 0 \implies t^2 - 6t + 8 = 0 \implies (t - 2)(t - 4) = 0$$

Divide by 3 and factorise.

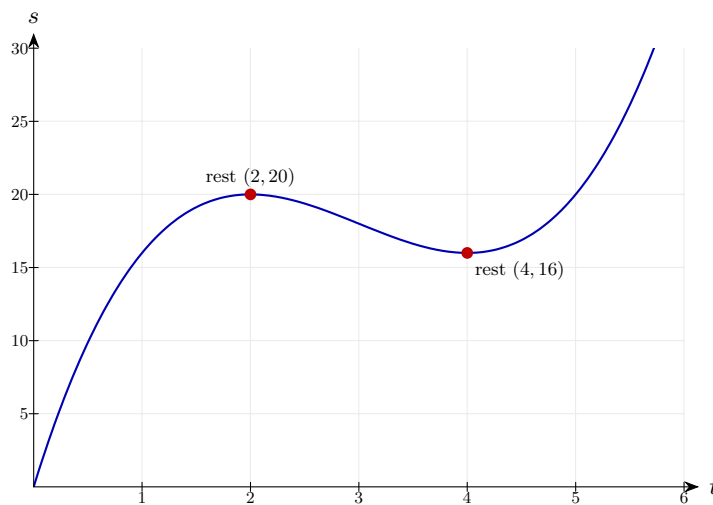
Step 3: Solve.

$$t = 2 \text{ s or } t = 4 \text{ s}$$

Both values are non-negative, so both are valid.

Final answer: (a) $v = 3t^2 - 18t + 24$
 (b) $t = 2 \text{ s}$ and $t = 4 \text{ s}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 05 (d5-005)

Difficulty: Foundation / Marks: 3

A motorbike starts from rest and travels along a straight road. After t seconds its displacement is $s = 3t^2$ metres. (a) Find the velocity after t seconds. (b) Find the velocity after 6 seconds. (c) Find the constant acceleration of the motorbike.

Worked solution.

Step 1: (a) Differentiate $s = 3t^2$.

$$v = \frac{ds}{dt} = 6t$$

Power rule.

Step 2: (b) Substitute $t = 6$.

$$v = 6(6) = 36 \text{ ms}^{-1}$$

Replace t with 6.

Step 3: (c) Differentiate $v = 6t$.

$$a = \frac{dv}{dt} = 6 \text{ ms}^{-2}$$

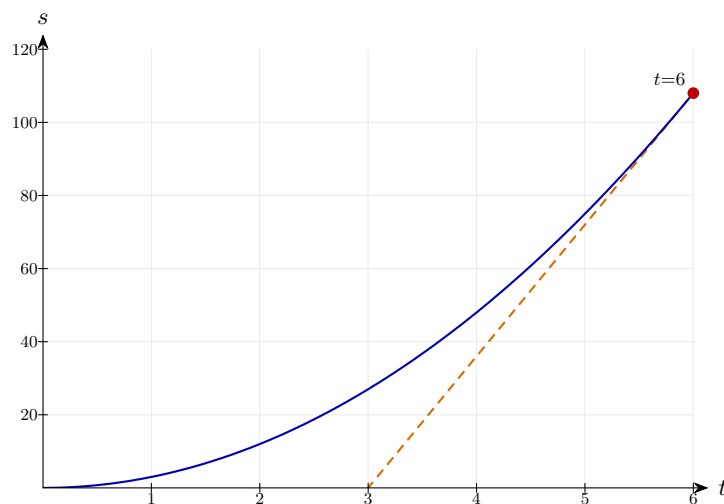
The acceleration is constant at 6 ms^{-2} .

Final answer: (a) $v = 6t$

(b) 36 ms^{-1}

(c) $a = 6 \text{ ms}^{-2}$

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 06 (d5-006)

Difficulty: Foundation / Marks: 5

A particle's displacement in metres is given by $s = 2t^3 - 15t^2 + 36t$ for $t \geq 0$. (a) Find the velocity when $t = 3$. (b) Find the acceleration when $t = 3$. (c) Find the value of t when the acceleration is zero.

Worked solution.

Step 1: Find velocity.

$$v = \frac{ds}{dt} = 6t^2 - 30t + 36$$

Differentiate s .

Step 2: (a) Substitute $t = 3$.

$$v(3) = 6(9) - 30(3) + 36 = 54 - 90 + 36 = 0 \text{ ms}^{-1}$$

The particle is momentarily at rest at $t = 3$.

Step 3: Find acceleration.

$$a = \frac{dv}{dt} = 12t - 30$$

Differentiate v .

Step 4: (b) Substitute $t = 3$.

$$a(3) = 12(3) - 30 = 36 - 30 = 6 \text{ ms}^{-2}$$

Replace t with 3.

Step 5: (c) Set $a = 0$.

$$12t - 30 = 0 \implies t = 2.5 \text{ s}$$

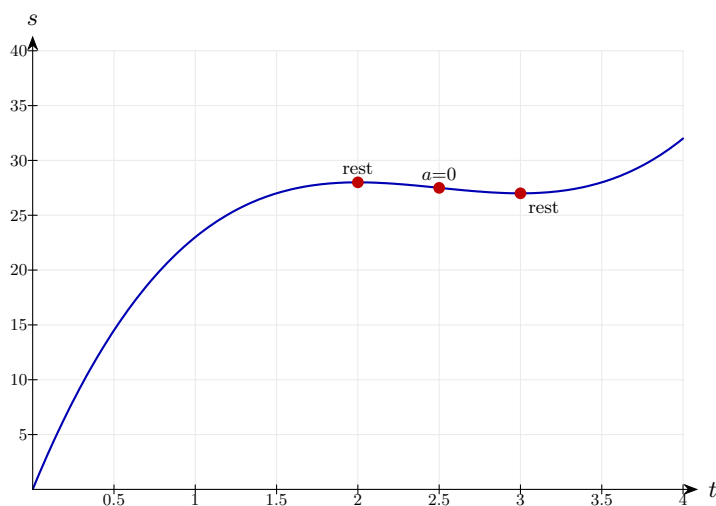
Solve the linear equation.

Final answer: (a) 0 ms^{-1}

(b) 6 ms^{-2}

(c) $t = 2.5 \text{ s}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 07 (d5-007)

Difficulty: Foundation / Marks: 4

A particle moves so that its displacement is $s = 5t - t^2$ metres for $0 \leq t \leq 5$. (a) Find the velocity function. (b) Find the time when the particle changes direction. (c) Find the maximum displacement from the starting point.

Worked solution.

Step 1: (a) Differentiate.

$$v = \frac{ds}{dt} = 5 - 2t$$

Power rule.

Step 2: (b) The particle changes direction when $v = 0$.

$$5 - 2t = 0 \implies t = 2.5 \text{ s}$$

Velocity changes sign here: positive before, negative after $\rightarrow$ particle reverses.

Step 3: (c) Maximum displacement at $t = 2.5$.

$$s = 5(2.5) - (2.5)^2 = 12.5 - 6.25 = 6.25 \text{ m}$$

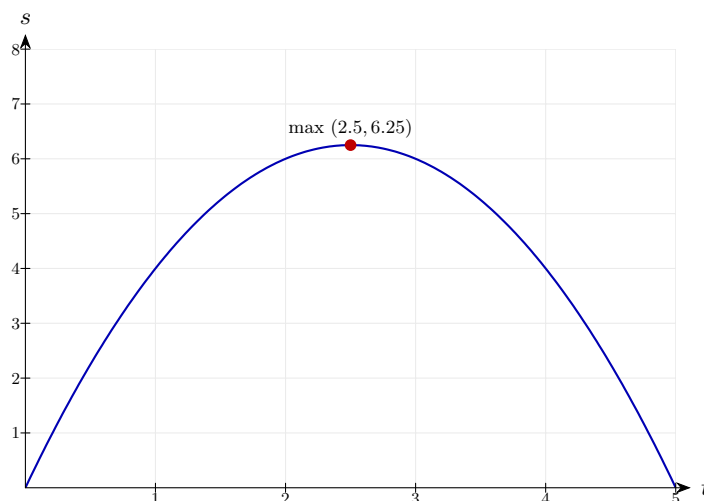
Substitute $t = 2.5$ into s .

Final answer: (a) $v = 5 - 2t$

(b) $t = 2.5 \text{ s}$

(c) Maximum displacement = 6.25 m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 08 (d5-008)

Difficulty: Foundation / Marks: 3

A ball is dropped from a height. Its velocity $v \text{ ms}^{-1}$ after t seconds is $v = 10t$. (a) Find the acceleration of the ball. (b) Find the time taken for the ball to reach 35 ms^{-1} .

Worked solution.

Step 1: (a) Differentiate v with respect to t .

$$a = \frac{dv}{dt} = 10 \text{ ms}^{-2}$$

The acceleration is constant — this approximates free-fall under gravity.

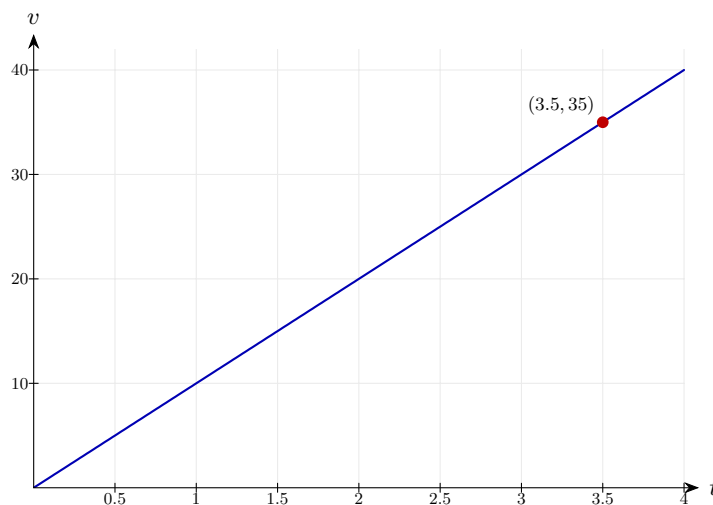
Step 2: (b) Set $v = 35$.

$$10t = 35 \implies t = 3.5 \text{ s}$$

Solve for t .

Final answer: (a) $a = 10 \text{ ms}^{-2}$
(b) $t = 3.5 \text{ s}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 09 (d5-009)

Difficulty: Foundation / Marks: 5

A particle's displacement is $s = t^3 - 4t^2 + 4t$ metres after t seconds. (a) Find the velocity function. (b) Find the times when the particle is at rest. (c) Find the acceleration when $t = 1$.

Worked solution.

Step 1: (a) Differentiate.

$$v = 3t^2 - 8t + 4$$

Power rule.

Step 2: (b) Set $v = 0$.

$$3t^2 - 8t + 4 = 0 \implies (3t - 2)(t - 2) = 0 \implies t = \frac{2}{3} \text{ or } t = 2$$

Factorise the quadratic.

Step 3: (c) Acceleration function.

$$a = \frac{dv}{dt} = 6t - 8$$

Differentiate v .

Step 4: Evaluate at $t = 1$.

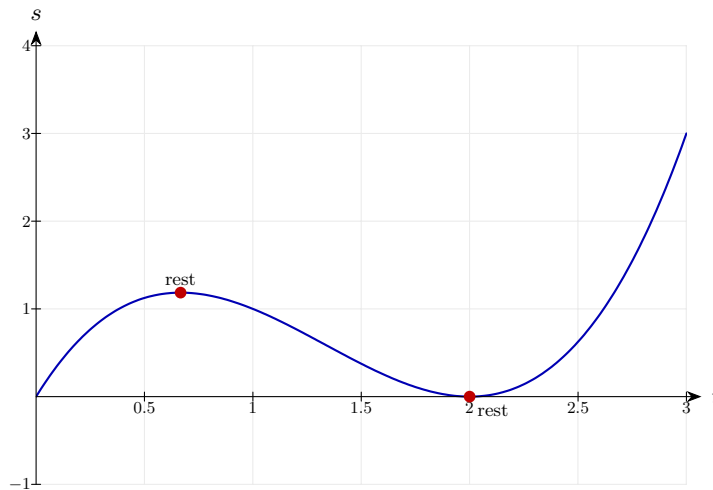
$$a(1) = 6(1) - 8 = -2 \text{ ms}^{-2}$$

The negative sign means the particle is decelerating at this instant.

Final answer: (a) $v = 3t^2 - 8t + 4$

- (b) $t = \frac{2}{3}$ s and $t = 2$ s
 (c) -2 ms^{-2}

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 10 (d5-010)

Difficulty: Foundation / Marks: 5

A rectangular garden has length x m and width y m. The perimeter of the garden is 60 m. (a) Write y in terms of x . (b) Write down an expression for the area A in terms of x only. (c) Use calculus to find the value of x that maximises the area, and find that maximum area.

Worked solution.

Step 1: (a) Use the perimeter condition.

$$2x + 2y = 60 \implies y = 30 - x$$

Rearrange the perimeter formula.

Step 2: (b) Area expression.

$$A = xy = x(30 - x) = 30x - x^2$$

Substitute $y = 30 - x$ into $A = xy$.

Step 3: (c) Differentiate and set to zero.

$$\frac{dA}{dx} = 30 - 2x = 0 \implies x = 15$$

Stationary point at $x = 15$.

Step 4: Confirm maximum.

$$\frac{d^2A}{dx^2} = -2 < 0 \implies \text{maximum}$$

Negative second derivative confirms a maximum.

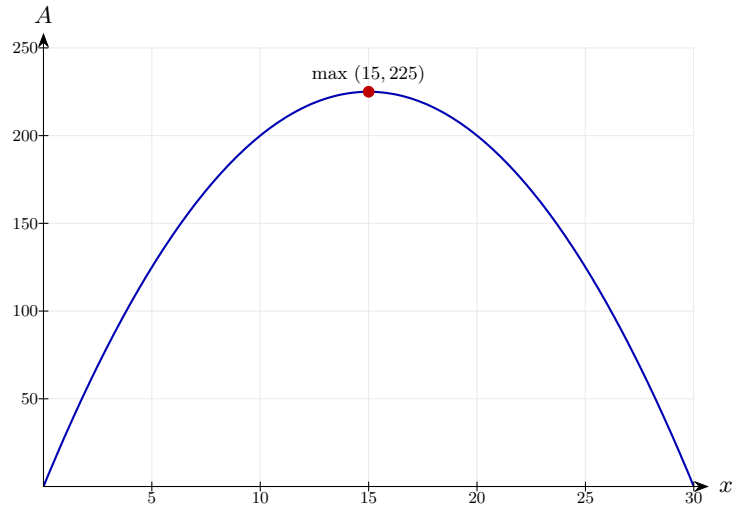
Step 5: Find maximum area.

$$A = 15 \times 15 = 225 \text{ m}^2$$

Substitute $x = 15$ and $y = 15$.

Final answer: (a) $y = 30 - x$
 (b) $A = 30x - x^2$
 (c) Maximum area = 225 m^2 when $x = 15$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 11 (d5-011)

Difficulty: Foundation / Marks: 5

A farmer uses 80 m of fencing to enclose a rectangular field on three sides (the fourth side is a wall). The width of the field is x m. (a) Show that the area of the field is $A = x(80 - 2x)$. (b) Find the maximum possible area using calculus.

Worked solution.

Step 1: (a) The three sides use $2x + l = 80$, where l is the length.

$$l = 80 - 2x \implies A = x \cdot l = x(80 - 2x)$$

Two widths x and one length l make up the 80 m of fencing.

Step 2: (b) Expand and differentiate.

$$A = 80x - 2x^2 \implies \frac{dA}{dx} = 80 - 4x$$

Power rule.

Step 3: Set to zero.

$$80 - 4x = 0 \implies x = 20$$

Stationary point at $x = 20$.

Step 4: Confirm maximum and find area.

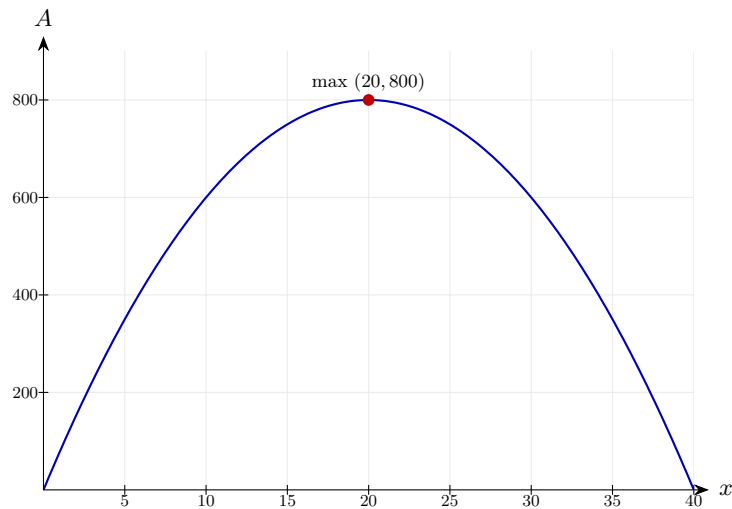
$$\frac{d^2A}{dx^2} = -4 < 0; \quad A = 20(80 - 40) = 20 \times 40 = 800 \text{ m}^2$$

Negative second derivative confirms maximum.

Final answer: (a) Shown.

(b) Maximum area = 800 m^2 when $x = 20$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 12 (d5-012)

Difficulty: Foundation / Marks: 5

A ball is thrown vertically upwards with an initial speed of 25 ms^{-1} . After t seconds its height is $h = 25t - 5t^2$ metres. (a) Find the height after 1 second. (b) Use calculus to find the maximum height. (c) Find the time when the ball returns to ground level.

Worked solution.

Step 1: (a) Substitute $t = 1$.

$$h(1) = 25 - 5 = 20 \text{ m}$$

Direct substitution.

Step 2: (b) Differentiate and set to zero.

$$\frac{dh}{dt} = 25 - 10t = 0 \implies t = 2.5 \text{ s}$$

Maximum height when velocity is zero.

Step 3: Find the maximum height.

$$h(2.5) = 25(2.5) - 5(6.25) = 62.5 - 31.25 = 31.25 \text{ m}$$

Substitute $t = 2.5$.

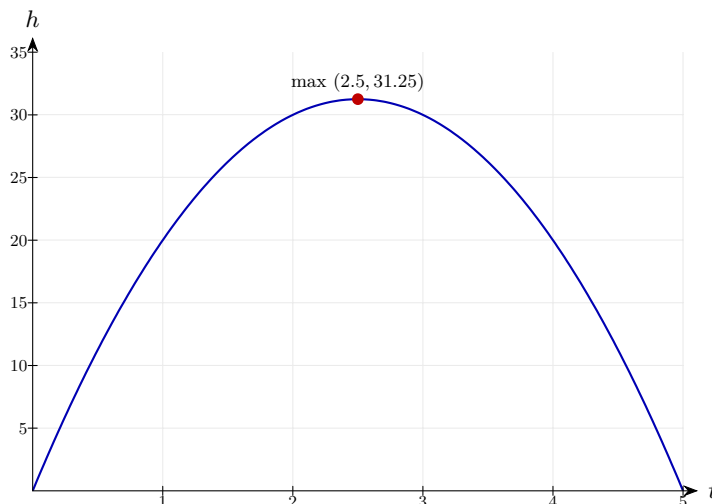
Step 4: (c) Set $h = 0$.

$$25t - 5t^2 = 0 \implies 5t(5 - t) = 0 \implies t = 0 \text{ or } t = 5$$

$t = 0$ is the launch; $t = 5$ is when it lands.

Final answer: (a) 20 m
 (b) 31.25 m at $t = 2.5$ s
 (c) $t = 5$ s

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 13 (d5-013)

Difficulty: Foundation / Marks: 5

A rectangular enclosure is to be made with 120 m of fencing, enclosing a field on all four sides. Find the dimensions that give the maximum area, and state that area.

Worked solution.

Step 1: Let the length be x and width be y . Perimeter: $2x + 2y = 120$.

$$y = 60 - x$$

Express y in terms of x .

Step 2: Area in terms of x .

$$A = xy = x(60 - x) = 60x - x^2$$

Substitute $y = 60 - x$.

Step 3: Differentiate and set to zero.

$$\frac{dA}{dx} = 60 - 2x = 0 \implies x = 30$$

Stationary point at $x = 30$.

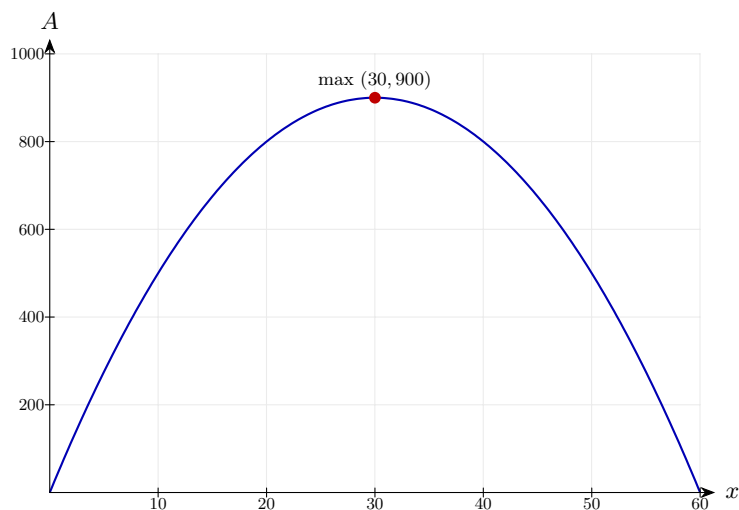
Step 4: Confirm maximum and find dimensions.

$$\frac{d^2A}{dx^2} = -2 < 0; \quad y = 30 \text{ m}; \quad A = 30^2 = 900 \text{ m}^2$$

Both sides equal 30 m — the optimal shape is a square.

Final answer: Length = width = 30 m; maximum area = 900 m²

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 14 (d5-014)

Difficulty: Foundation / Marks: 6

A child cuts equal squares of side x cm from each corner of a 30×30 cm piece of card and folds the sides up to make an open box. (a) Show that the volume of the box is $V = x(30 - 2x)^2$. (b) Find the value of x that maximises the volume.

Worked solution.

Step 1: (a) After cutting, the base has side $(30 - 2x)$ cm and the height is x cm.

$$V = x(30 - 2x)^2$$

Volume = base area $\times$ height; the base is a square of side $(30 - 2x)$.

Step 2: (b) Expand.

$$V = x(900 - 120x + 4x^2) = 900x - 120x^2 + 4x^3$$

Expand $(30 - 2x)^2$ first, then multiply by x .

Step 3: Differentiate.

$$\frac{dV}{dx} = 900 - 240x + 12x^2$$

Power rule.

Step 4: Set to zero and simplify.

$$12x^2 - 240x + 900 = 0 \implies x^2 - 20x + 75 = 0 \implies (x - 5)(x - 15) = 0$$

Divide by 12, then factorise.

Step 5: Identify the valid solution.

$$x = 5 \text{ or } x = 15; \quad x = 15 \text{ gives } (30 - 30) = 0, \text{ so reject}$$

$x = 15$ gives zero base length — not a valid box. $x = 5$ gives a real box.

Step 6: Confirm maximum at $x = 5$.

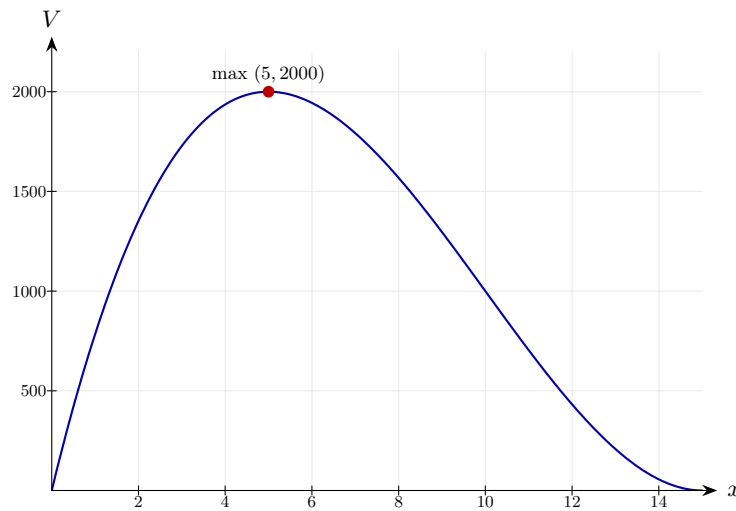
$$\frac{d^2V}{dx^2} = -240 + 24x; \quad x = 5 : -240 + 120 = -120 < 0 \Rightarrow \text{maximum}$$

Negative second derivative confirms a maximum.

Final answer: (a) Shown.

(b) $x = 5$ cm maximises the volume.

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 15 (d5-015)

Difficulty: Foundation / Marks: 6

A closed cylindrical can has radius r cm and height h cm. Its volume is 250π cm³. (a) Show that the total surface area is $A = 2\pi r^2 + \frac{500\pi}{r}$. (b) Find the value of r that minimises A .

Worked solution.

Step 1: (a) Volume: $\pi r^2 h = 250\pi$.

$$h = \frac{250}{r^2}$$

Rearrange to express h in terms of r .

Step 2: Surface area: two circles + curved surface.

$$A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \cdot \frac{250}{r^2} = 2\pi r^2 + \frac{500\pi}{r}$$

Substitute $h = \frac{250}{r^2}$ into the surface area formula.

Step 3: (b) Rewrite and differentiate.

$$A = 2\pi r^2 + 500\pi r^{-1} \implies \frac{dA}{dr} = 4\pi r - \frac{500\pi}{r^2}$$

Convert $\frac{1}{r}$ to r^{-1} before differentiating.

Step 4: Set to zero.

$$4\pi r = \frac{500\pi}{r^2} \implies 4r^3 = 500 \implies r^3 = 125 \implies r = 5 \text{ cm}$$

Multiply both sides by r^2 and solve for r .

Step 5: Confirm minimum.

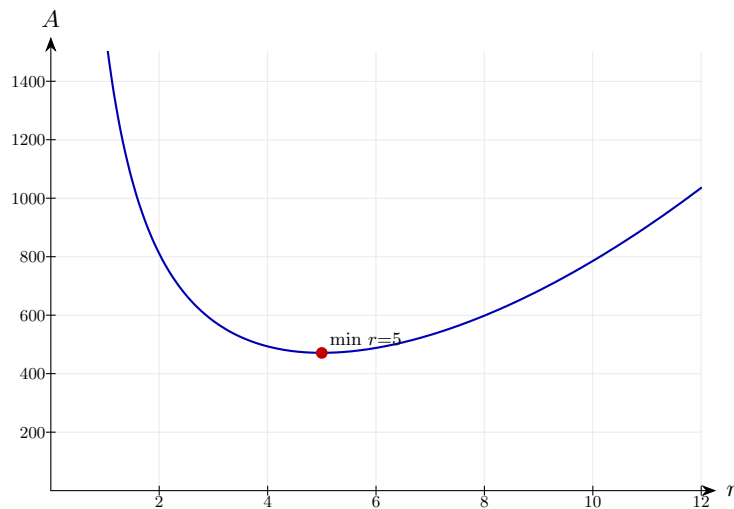
$$\frac{d^2 A}{dr^2} = 4\pi + \frac{1000\pi}{r^3}; \quad r = 5 : 4\pi + 8\pi = 12\pi > 0 \Rightarrow \text{minimum}$$

Positive second derivative confirms a minimum.

Final answer: (a) Shown.

(b) $r = 5 \text{ cm}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 16 (d5-016)

Difficulty: Foundation / Marks: 5

A particle moves so that its velocity at time t seconds is $v = 3t^2 - 12t + 9 \text{ ms}^{-1}$. (a) Find the acceleration function. (b) Find the acceleration when the particle has zero velocity. (c) Find the minimum velocity.

Worked solution.

Step 1: (a) Differentiate v with respect to t .

$$a = \frac{dv}{dt} = 6t - 12$$

Acceleration is the derivative of velocity.

Step 2: (b) Find when $v = 0$.

$$3t^2 - 12t + 9 = 0 \implies (t - 1)(t - 3) = 0 \implies t = 1 \text{ or } t = 3$$

Divide by 3 and factorise.

Step 3: Acceleration at $t = 1$ and $t = 3$.

$$a(1) = 6 - 12 = -6 \text{ ms}^{-2}; \quad a(3) = 18 - 12 = 6 \text{ ms}^{-2}$$

Substitute each t into a .

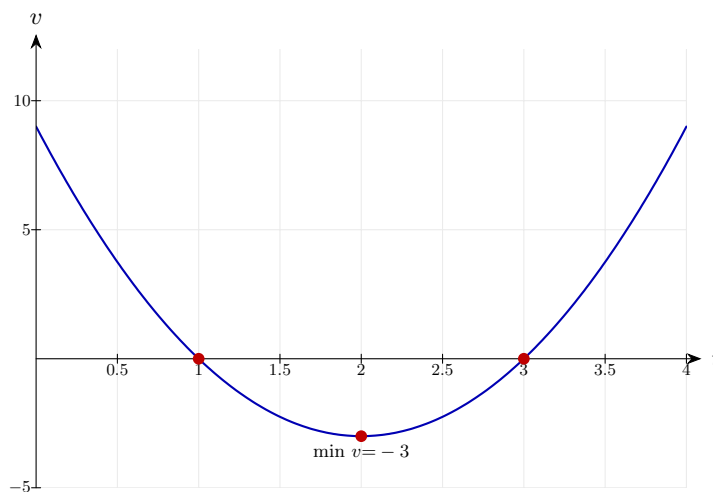
Step 4: (c) Minimum velocity: set $\frac{dv}{dt} = 0$.

$$6t - 12 = 0 \implies t = 2; \quad v(2) = 12 - 24 + 9 = -3 \text{ ms}^{-1}$$

Find the turning point of v . The negative value means the particle moves in reverse.

Final answer: (a) $a = 6t - 12$
(b) -6 ms^{-2} at $t = 1$; 6 ms^{-2} at $t = 3$
(c) Minimum velocity $= -3 \text{ ms}^{-1}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 17 (d5-017)

Difficulty: Foundation / Marks: 6

A square piece of metal of side 24 cm has squares of side x cm cut from each corner and is then folded into an open tray. (a) Write down an expression for the volume V of the tray. (b) Find the value of x that gives the maximum volume, and state this maximum volume.

Worked solution.

Step 1: (a) After cutting, the base side is $(24 - 2x)$ and the height is x .

$$V = x(24 - 2x)^2$$

Volume = base area $\times$ height.

Step 2: Expand.

$$V = x(576 - 96x + 4x^2) = 576x - 96x^2 + 4x^3$$

Expand the bracket.

Step 3: Differentiate.

$$\frac{dV}{dx} = 576 - 192x + 12x^2$$

Power rule.

Step 4: Set to zero and solve.

$$12x^2 - 192x + 576 = 0 \implies x^2 - 16x + 48 = 0 \implies (x - 4)(x - 12) = 0$$

Divide by 12 and factorise.

Step 5: Reject invalid root.

$$x = 4 \text{ or } x = 12; \quad x = 12 \text{ gives zero base - reject}$$

$x = 12$ gives $24 - 24 = 0$ base — not a valid tray.

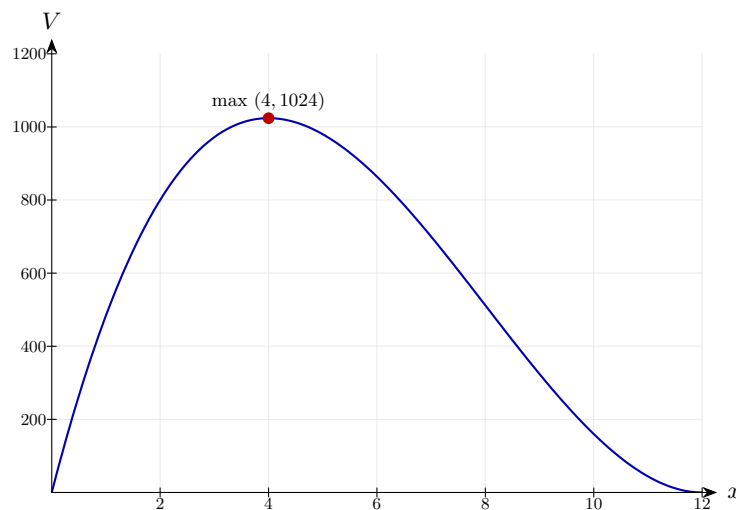
Step 6: Maximum volume at $x = 4$.

$$V = 4(24 - 8)^2 = 4 \times 256 = 1024 \text{ cm}^3$$

Substitute $x = 4$ and confirm it is a maximum (second derivative is negative).

Final answer: (a) $V = x(24 - 2x)^2$
(b) $x = 4$ cm; maximum volume = 1024 cm^3

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 18 (d5-018)

Difficulty: Foundation / Marks: 6

A rectangular sheet of cardboard measures 20 cm by 12 cm. Equal squares of side x cm are cut from each corner and the sides folded up to form an open box. (a) Show that the volume is $V = 4x^3 - 64x^2 + 240x$. (b) Find the value of x that maximises the volume.

Worked solution.

Step 1: (a) After cutting, the box has dimensions $(20 - 2x) \times (12 - 2x) \times x$.

$$V = x(20 - 2x)(12 - 2x)$$

Length, width and height of the open box.

Step 2: Expand.

$$V = x(240 - 40x - 24x + 4x^2) = x(240 - 64x + 4x^2) = 240x - 64x^2 + 4x^3$$

Multiply out and simplify.

Step 3: (b) Differentiate.

$$\frac{dV}{dx} = 240 - 128x + 12x^2$$

Power rule.

Step 4: Set to zero.

$$12x^2 - 128x + 240 = 0 \implies 3x^2 - 32x + 60 = 0$$

Divide by 4 to simplify. This quadratic does not factorise over the integers (testing $x = 2$: $12 - 64 + 60 = 8 \neq 0$), so the quadratic formula is needed.

Step 5: Use the quadratic formula.

$$x = \frac{32 \pm \sqrt{1024 - 720}}{6} = \frac{32 \pm \sqrt{304}}{6} \approx \frac{32 \pm 17.44}{6}$$

$$a = 12, b = -128, c = 240.$$

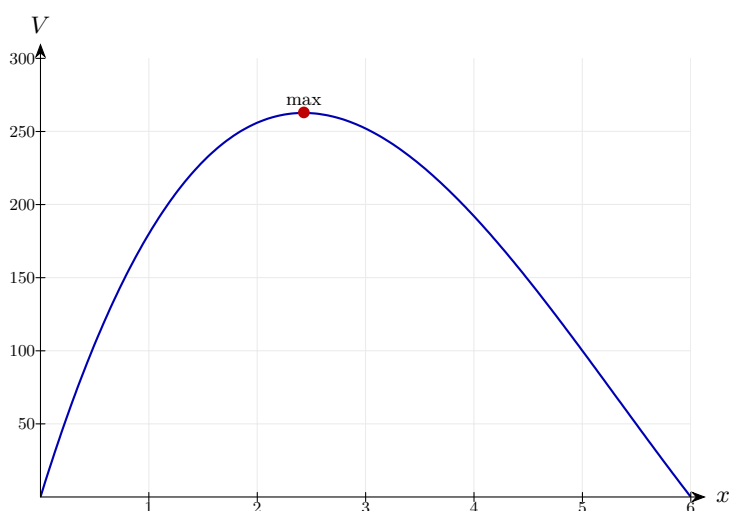
Step 6: Select valid solution.

$$x \approx \frac{32 - 17.44}{6} \approx 2.43 \text{ cm (to 3 s.f.); } x \approx 8.24 \text{ rejected (} > 6 \text{)}$$

The maximum allowed x is half the shortest side: $\frac{12}{2} = 6$ cm. So $x \approx 8.24$ is rejected.

Final answer: (a) Shown: $V = 4x^3 - 64x^2 + 240x$
(b) $x \approx 2.43$ cm

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 19 (d5-019)

Difficulty: Foundation / Marks: 7

A cylindrical storage tank (open at the top) has radius r m and height h m. The volume must be 16π m³. (a) Express h in terms of r . (b) Show that the surface area (base + curved side) is $A = \pi r^2 + \frac{32\pi}{r}$. (c) Find the value of r that minimises the surface area.

Worked solution.

Step 1: (a) Use the volume formula.

$$\pi r^2 h = 16\pi \implies h = \frac{16}{r^2}$$

Rearrange to make h the subject.

Step 2: (b) Surface area = base + curved side.

$$A = \pi r^2 + 2\pi r h = \pi r^2 + 2\pi r \cdot \frac{16}{r^2} = \pi r^2 + \frac{32\pi}{r}$$

Substitute $h = \frac{16}{r^2}$.

Step 3: (c) Differentiate.

$$\frac{dA}{dr} = 2\pi r - \frac{32\pi}{r^2}$$

$$\frac{d}{dr} \left(\frac{32\pi}{r} \right) = -\frac{32\pi}{r^2}.$$

Step 4: Set to zero.

$$2\pi r = \frac{32\pi}{r^2} \implies 2r^3 = 32 \implies r^3 = 16 \implies r = 2\sqrt[3]{2} \approx 2.52 \text{ m}$$

Multiply by r^2 and solve for r .

Step 5: Confirm minimum.

$$\frac{d^2 A}{dr^2} = 2\pi + \frac{64\pi}{r^3} > 0 \Rightarrow \text{minimum}$$

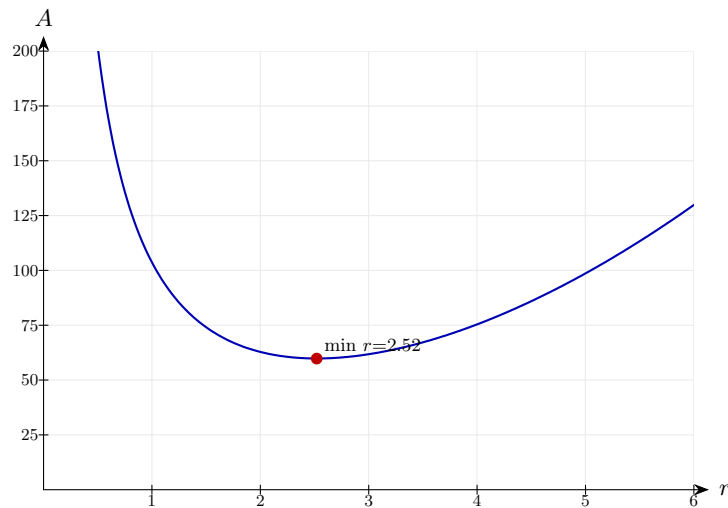
Both terms are positive, so the second derivative is always positive.

Final answer: (a) $h = \frac{16}{r^2}$

(b) Shown.

(c) $r = 2\sqrt[3]{2} \approx 2.52$ m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 20 (d5-020)

Difficulty: Foundation / Marks: 6

A particle's displacement is $s = t^4 - 8t^2$ metres after t seconds for $t \geq 0$. (a) Find the velocity and acceleration functions. (b) Find the values of t when the particle is at rest. (c) Find the acceleration at each of those times.

Worked solution.

Step 1: (a) Velocity.

$$v = \frac{ds}{dt} = 4t^3 - 16t$$

Differentiate s .

Step 2: Acceleration.

$$a = \frac{dv}{dt} = 12t^2 - 16$$

Differentiate v .

Step 3: (b) Set $v = 0$.

$$4t^3 - 16t = 0 \implies 4t(t^2 - 4) = 0 \implies t = 0, t = 2 \text{ (} t = -2 \text{ rejected)}$$

For $t \geq 0$, only $t = 0$ and $t = 2$ are valid.

Step 4: (c) Acceleration at rest times.

$$a(0) = -16 \text{ ms}^{-2}; \quad a(2) = 12(4) - 16 = 32 \text{ ms}^{-2}$$

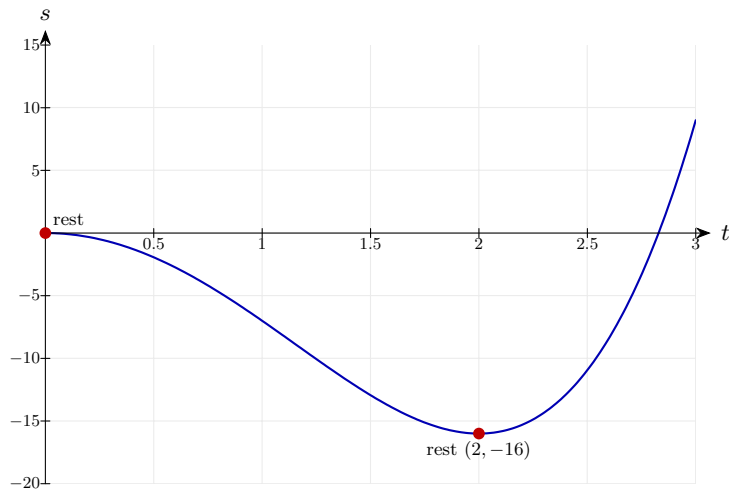
Substitute $t = 0$ and $t = 2$ into $a = 12t^2 - 16$.

Final answer: (a) $v = 4t^3 - 16t$, $a = 12t^2 - 16$

(b) $t = 0$ and $t = 2$ s

(c) -16 ms^{-2} and 32 ms^{-2}

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 21 (d5-021)

Difficulty: Foundation / Marks: 5

A wire of length 48 cm is bent to form a rectangle. Find the dimensions that maximise the area enclosed, and state that maximum area.

Worked solution.

Step 1: Let the length be l and width w . Perimeter = $2l + 2w = 48$.

$$l = 24 - w$$

Express one variable in terms of the other.

Step 2: Area.

$$A = lw = w(24 - w) = 24w - w^2$$

Substitute $l = 24 - w$.

Step 3: Differentiate and set to zero.

$$\frac{dA}{dw} = 24 - 2w = 0 \implies w = 12 \text{ cm}$$

Maximum area stationary point.

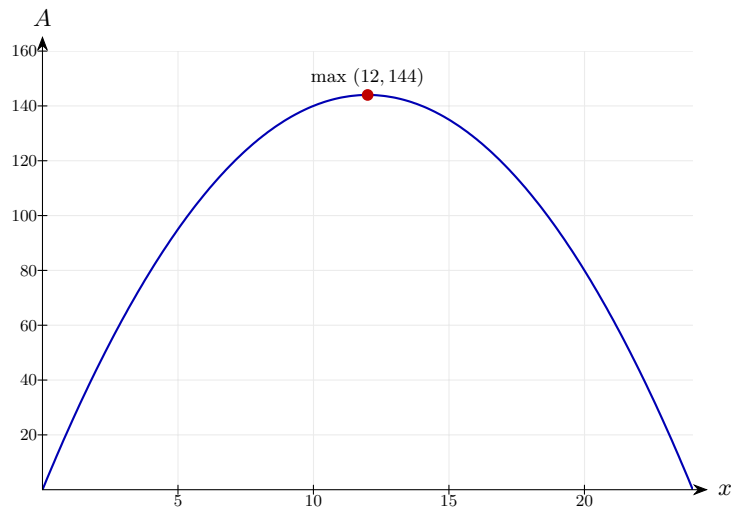
Step 4: Confirm maximum and state dimensions.

$$l = 24 - 12 = 12 \text{ cm}; \quad A = 12 \times 12 = 144 \text{ cm}^2$$

A square always gives the maximum area for a fixed perimeter.

Final answer: Dimensions: 12 cm $\times$ 12 cm; maximum area = 144 cm²

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 22 (d5-022)

Difficulty: Foundation / Marks: 6

A rectangular pen is built against a long straight wall (the wall forms one side). The pen uses 100 m of fencing for the remaining three sides, and the width of the pen (perpendicular to the wall) is x m. (a) Write down the length of the pen in terms of x . (b) Show that the area is $A = 100x - 2x^2$. (c) Find the maximum area and the dimensions of the pen.

Worked solution.

Step 1: (a) Three sides: two widths x and one length l . So $2x + l = 100$.

$$l = 100 - 2x$$

Rearrange to express l in terms of x .

Step 2: (b) Area.

$$A = xl = x(100 - 2x) = 100x - 2x^2$$

Substitute $l = 100 - 2x$.

Step 3: (c) Differentiate and set to zero.

$$\frac{dA}{dx} = 100 - 4x = 0 \implies x = 25 \text{ m}$$

Stationary point.

Step 4: Confirm maximum.

$$\frac{d^2A}{dx^2} = -4 < 0 \implies \text{maximum}$$

Negative second derivative.

Step 5: Maximum area.

$$l = 100 - 50 = 50 \text{ m}; \quad A = 25 \times 50 = 1250 \text{ m}^2$$

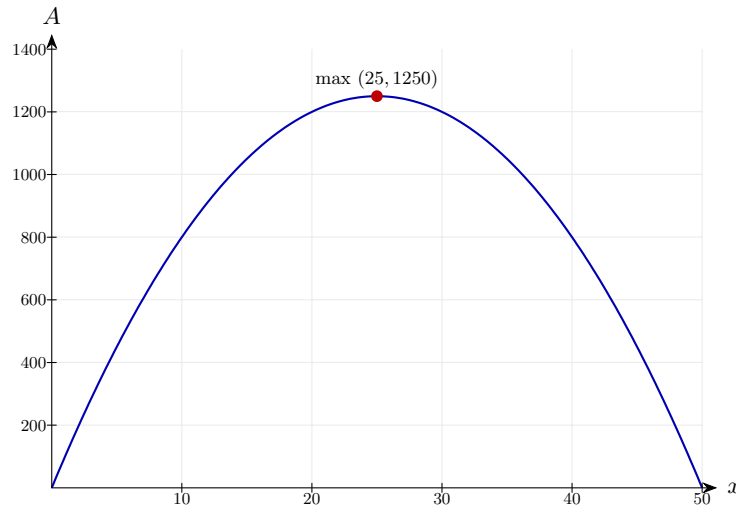
$x = 25$ m and $l = 50$ m.

Final answer: (a) $l = 100 - 2x$

(b) Shown.

(c) Width = 25 m, length = 50 m; max area = 1250 m²

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 23 (d5-023)

Difficulty: Foundation / Marks: 5

A model rocket's height in metres after t seconds is $h = -2t^3 + 18t^2 + 5$ for $0 \leq t \leq 10$.

(a) Find the velocity $\frac{dh}{dt}$. (b) Find the times when the rocket is neither ascending nor descending. (c) Find the maximum height.

Worked solution.

Step 1: (a) Differentiate.

$$\frac{dh}{dt} = -6t^2 + 36t$$

Power rule.

Step 2: (b) Set velocity to zero.

$$-6t^2 + 36t = 0 \implies -6t(t - 6) = 0 \implies t = 0 \text{ or } t = 6$$

Factorise.

Step 3: (c) Maximum height at $t = 6$.

$$h(6) = -2(216) + 18(36) + 5 = -432 + 648 + 5 = 221 \text{ m}$$

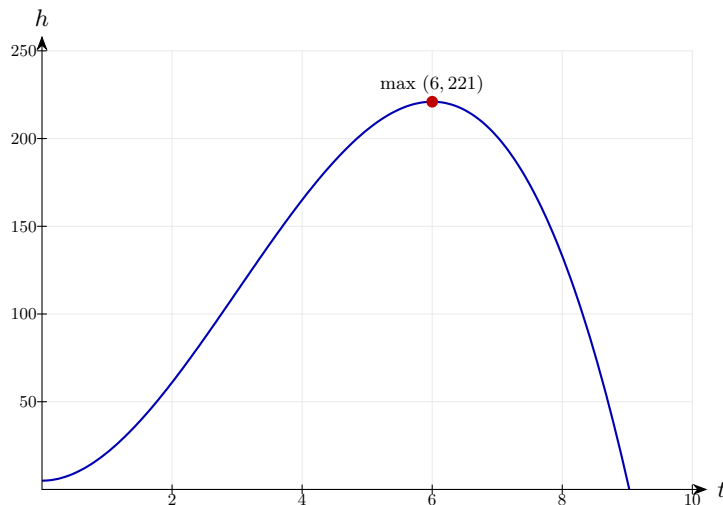
Substitute $t = 6$ into h . ($t = 0$ is launch, not a maximum.)

Final answer: (a) $v = -6t^2 + 36t$

(b) $t = 0$ and $t = 6$ s

(c) Maximum height = 221 m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 24 (d5-024)

Difficulty: Foundation / Marks: 6

A piece of wire 60 cm long is cut into two pieces. One piece is bent into a square of side x cm and the other into a rectangle of dimensions $2x$ cm by y cm. (a) Write an expression for y in terms of x . (b) Find the total area A of both shapes in terms of x . (c) Use calculus to find the value of x that minimises the total area.

Worked solution.

Step 1: (a) Total wire: perimeter of square + perimeter of rectangle = 60.

$$4x + 2(2x + y) = 60 \implies 4x + 4x + 2y = 60 \implies y = 30 - 4x$$

Perimeter of rectangle = $2(2x + y)$.

Step 2: (b) Total area.

$$A = x^2 + 2x \cdot y = x^2 + 2x(30 - 4x) = x^2 + 60x - 8x^2 = -7x^2 + 60x$$

Area of square = x^2 ; area of rectangle = $2xy$.

Step 3: (c) Differentiate.

$$\frac{dA}{dx} = -14x + 60 = 0 \implies x = \frac{60}{14} = \frac{30}{7} \approx 4.29 \text{ cm}$$

Set derivative to zero and solve.

Step 4: Confirm minimum.

$$\frac{d^2A}{dx^2} = -14 < 0$$

Wait — negative second derivative means this is a maximum, not a minimum! The area function is a downward parabola, so it has a maximum at $x = \frac{30}{7}$. For a minimum of a downward parabola on a

restricted domain, check the endpoints. x must satisfy $y = 30 - 4x \geq 0$, so $x \leq 7.5$, and $x > 0$. The minimum area occurs at an endpoint of the domain.

Step 5: Check endpoints.

$$A(0) = 0; \quad A(7.5) = -7(56.25) + 60(7.5) = -393.75 + 450 = 56.25 \text{ cm}^2$$

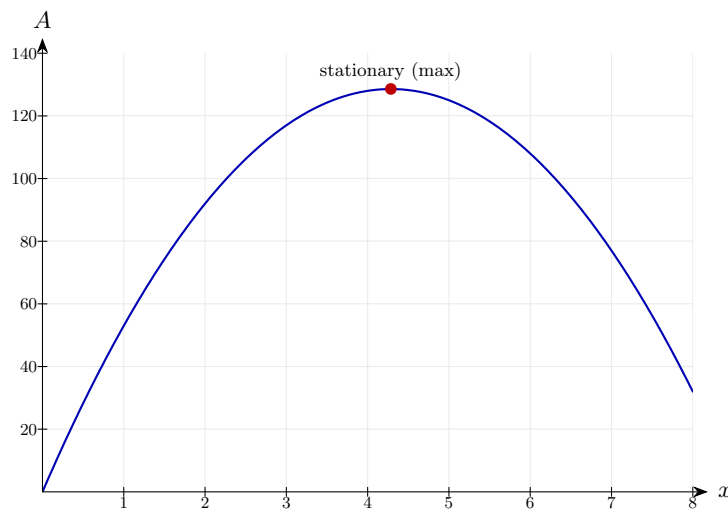
The minimum total area is 0 cm^2 (degenerate case at $x = 0$). Minimum non-degenerate area is at $x \rightarrow 0^+$. The stationary point $x \approx 4.29$ gives the maximum area of about 128.6 cm^2 .

Final answer: (a) $y = 30 - 4x$

(b) $A = -7x^2 + 60x$

(c) The stationary point $x = \frac{30}{7}$ is a maximum ($\approx 128.6 \text{ cm}^2$); area is minimised at the domain boundaries.

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 25 (d5-025)

Difficulty: Foundation / Marks: 7

A closed rectangular box has a square base of side x cm and height h cm. The total surface area is 150 cm^2 . (a) Show that the volume is $V = \frac{150x - 2x^3}{4}$. (b) Find the value of x that maximises the volume.

Worked solution.

Step 1: (a) Surface area: 2 square faces + 4 rectangular faces.

$$2x^2 + 4xh = 150 \implies h = \frac{150 - 2x^2}{4x}$$

Rearrange to express h in terms of x .

Step 2: Volume.

$$V = x^2h = x^2 \cdot \frac{150 - 2x^2}{4x} = \frac{x(150 - 2x^2)}{4} = \frac{150x - 2x^3}{4}$$

Substitute h into $V = x^2h$.

Step 3: (b) Differentiate.

$$\frac{dV}{dx} = \frac{150 - 6x^2}{4}$$

Power rule applied inside the fraction.

Step 4: Set to zero.

$$150 - 6x^2 = 0 \implies x^2 = 25 \implies x = 5 \text{ cm}$$

Take the positive root since $x > 0$.

Step 5: Confirm maximum.

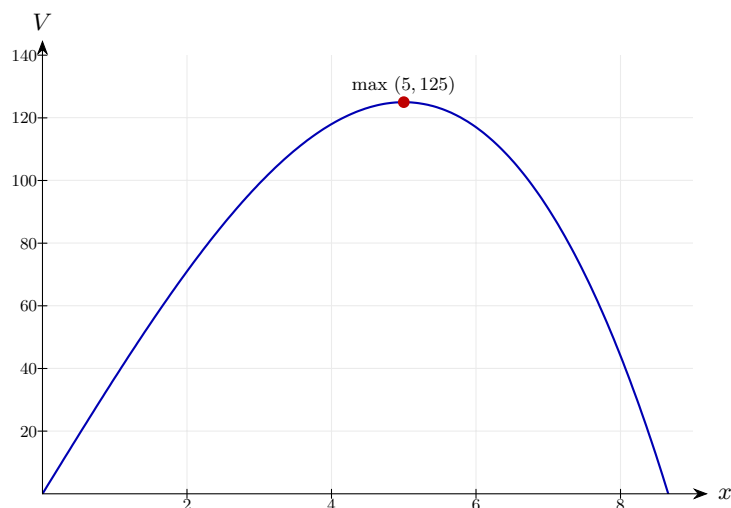
$$\frac{d^2V}{dx^2} = \frac{-12x}{4} = -3x; \quad x = 5 : -15 < 0 \Rightarrow \text{maximum}$$

Negative second derivative at $x = 5$.

Final answer: (a) Shown.

(b) $x = 5 \text{ cm}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 26 (d5-026)

Difficulty: Foundation / Marks: 6

A particle moves along a straight line. The displacement from the origin after t seconds is $s = 6t - t^2$ m for $0 \leq t \leq 8$. (a) Find the velocity when $t = 1$. (b) Find the displacement at the instant when the particle is momentarily at rest. (c) Find the total distance travelled in the first 8 seconds.

Worked solution.

Step 1: (a) Differentiate.

$$v = \frac{ds}{dt} = 6 - 2t; \quad v(1) = 6 - 2 = 4 \text{ ms}^{-1}$$

Power rule, then substitute $t = 1$.

Step 2: (b) Particle at rest: $v = 0$.

$$6 - 2t = 0 \implies t = 3 \text{ s}; \quad s(3) = 18 - 9 = 9 \text{ m}$$

Substitute $t = 3$ into s .

Step 3: (c) After $t = 3$, the particle reverses. Find $s(8)$.

$$s(8) = 48 - 64 = -16 \text{ m}$$

The particle is now 16 m on the other side of the origin.

Step 4: Total distance.

$$d = 9 + (9 - (-16)) = 9 + 25 = 34 \text{ m}$$

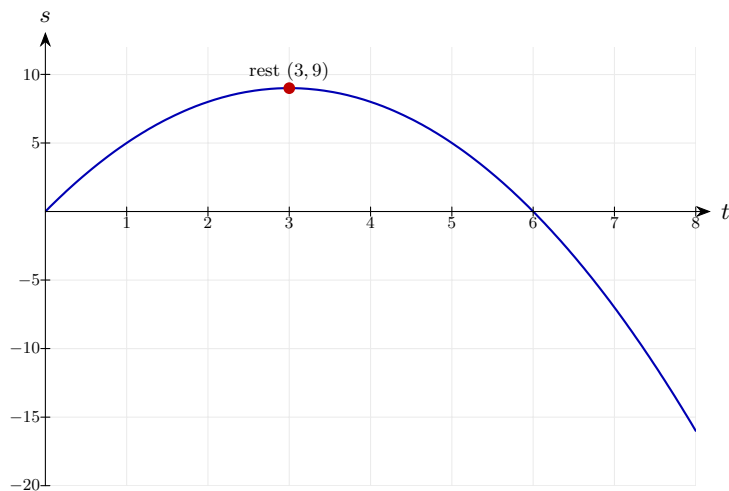
Outward journey: 9 m. Return journey: from $s = 9$ to $s = -16$ is 25 m.

Final answer: (a) 4 ms^{-1}

(b) 9 m from origin

(c) Total distance = 34 m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 27 (d5-027)

Difficulty: Foundation / Marks: 7

An open-topped cylindrical bucket is to be made from a fixed area of metal $300\pi \text{ cm}^2$. The radius is $r \text{ cm}$ and the height is $h \text{ cm}$. (a) Show that $h = \frac{300 - r^2}{2r}$. (b) Show that the volume is $V = \frac{\pi r}{2}(300 - r^2)$. (c) Find the value of r that maximises the volume.

Worked solution.

Step 1: (a) Surface area (open top): base + curved side.

$$\pi r^2 + 2\pi r h = 300\pi \implies r^2 + 2rh = 300 \implies h = \frac{300 - r^2}{2r}$$

Rearrange to make h the subject.

Step 2: (b) Volume.

$$V = \pi r^2 h = \pi r^2 \cdot \frac{300 - r^2}{2r} = \frac{\pi r(300 - r^2)}{2}$$

Substitute h into $V = \pi r^2 h$ and simplify.

Step 3: (c) Expand and differentiate.

$$V = \frac{\pi}{2}(300r - r^3) \implies \frac{dV}{dr} = \frac{\pi}{2}(300 - 3r^2)$$

Power rule applied inside the bracket.

Step 4: Set to zero.

$$300 - 3r^2 = 0 \implies r^2 = 100 \implies r = 10 \text{ cm}$$

Take the positive root.

Step 5: Confirm maximum.

$$\frac{d^2V}{dr^2} = \frac{\pi}{2}(-6r); \quad r = 10 : -30\pi < 0 \Rightarrow \text{maximum}$$

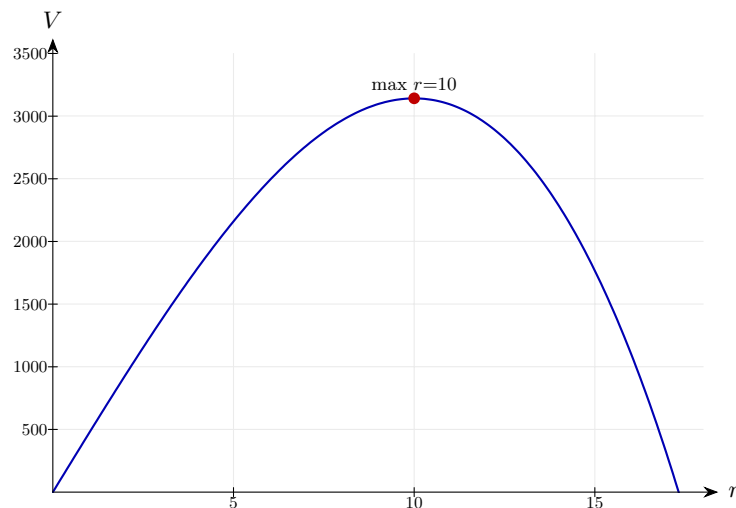
Negative second derivative.

Final answer: (a) Shown.

(b) Shown.

(c) $r = 10 \text{ cm}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 28 (d5-028)

Difficulty: Foundation / Marks: 6

A particle moves along the x -axis. Its displacement is $x = t^3 - 6t^2 + 12t - 3$ metres at time t seconds. (a) Find the velocity and acceleration functions. (b) Find the time when the acceleration is zero. (c) Show that the velocity is always positive for $t \geq 0$.

Worked solution.

Step 1: (a) Velocity.

$$v = \frac{dx}{dt} = 3t^2 - 12t + 12$$

Power rule.

Step 2: Acceleration.

$$a = \frac{dv}{dt} = 6t - 12$$

Differentiate v .

Step 3: (b) Set $a = 0$.

$$6t - 12 = 0 \implies t = 2 \text{ s}$$

Solve the linear equation.

Step 4: (c) Factorise or complete the square for v .

$$v = 3t^2 - 12t + 12 = 3(t^2 - 4t + 4) = 3(t - 2)^2$$

Recognise the perfect square.

Step 5: Conclude.

$$v = 3(t - 2)^2 \geq 0 \text{ for all } t$$

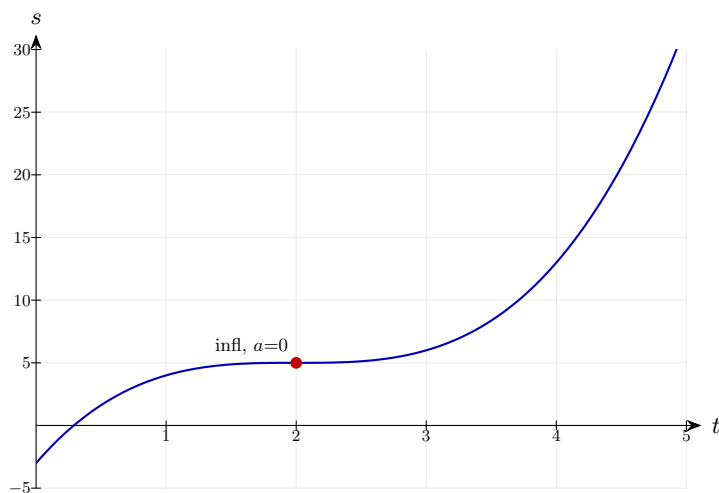
A square is always ≥ 0 , so velocity is never negative. At $t = 2$ it equals zero momentarily, but is positive otherwise.

Final answer: (a) $v = 3t^2 - 12t + 12$, $a = 6t - 12$

(b) $t = 2 \text{ s}$

(c) $v = 3(t - 2)^2 \geq 0$ for all $t \geq 0$ ✓

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 29 (d5-029)

Difficulty: Foundation / Marks: 6

A farmer needs to fence a rectangular area of 200 m^2 on all four sides. (a) Let the width be $x \text{ m}$. Write down the length in terms of x . (b) Find an expression for the total length of fencing needed. (c) Use calculus to find the dimensions that minimise the fencing used.

Worked solution.

Step 1: (a) Area = $x \times l = 200$.

$$l = \frac{200}{x}$$

Rearrange the area formula.

Step 2: (b) Total fencing.

$$P = 2x + 2l = 2x + \frac{400}{x}$$

Substitute $l = \frac{200}{x}$ into the perimeter formula.

Step 3: (c) Differentiate.

$$\frac{dP}{dx} = 2 - \frac{400}{x^2}$$

$$\frac{d}{dx} \left(\frac{400}{x} \right) = -\frac{400}{x^2}.$$

Step 4: Set to zero.

$$2 = \frac{400}{x^2} \implies x^2 = 200 \implies x = 10\sqrt{2} \approx 14.1 \text{ m}$$

Solve for x .

Step 5: Confirm minimum and find length.

$$\frac{d^2P}{dx^2} = \frac{800}{x^3} > 0 \implies \text{minimum}; \quad l = \frac{200}{10\sqrt{2}} = 10\sqrt{2}$$

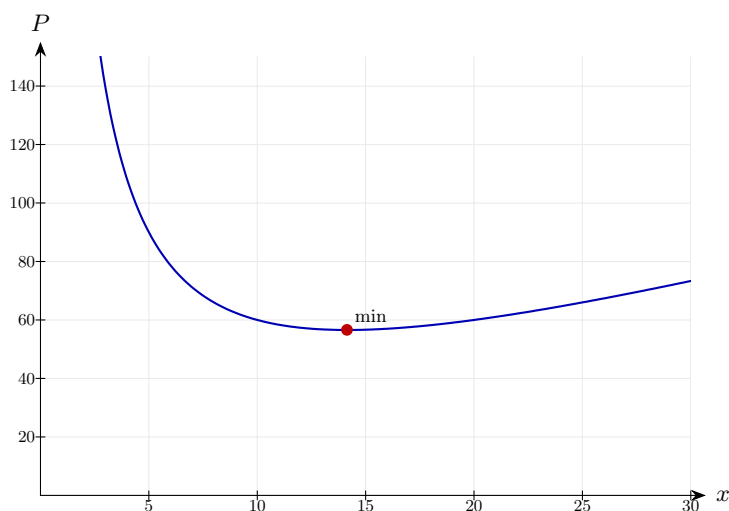
The optimal shape is again a square.

Final answer: (a) $l = \frac{200}{x}$

(b) $P = 2x + \frac{400}{x}$

(c) Both dimensions = $10\sqrt{2} \approx 14.1 \text{ m}$; minimum fencing = $40\sqrt{2} \approx 56.6 \text{ m}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 30 (d5-030)

Difficulty: Foundation / Marks: 6

A particle travels along a straight line. Its displacement s metres from a fixed point O after t seconds is $s = t^3 - 3t^2 - 9t + 2$ for $t \geq 0$. (a) Find the velocity function and hence the times when the particle passes through O travelling in the positive direction. (b) Find the acceleration when $t = 4$.

Worked solution.

Step 1: (a) Velocity.

$$v = 3t^2 - 6t - 9 = 3(t^2 - 2t - 3) = 3(t - 3)(t + 1)$$

Differentiate and factorise.

Step 2: Particle travels in positive direction when $v > 0$.

$$3(t - 3)(t + 1) > 0 \implies t > 3 \text{ (since } t \geq 0\text{)}$$

For $t \geq 0$: $(t + 1) > 0$ always, so need $(t - 3) > 0$.

Step 3: Find when the particle is at O : $s = 0$.

$$t^3 - 3t^2 - 9t + 2 = 0$$

This cubic has no simple rational roots — find the time when the particle passes through O travelling in the positive direction numerically or note that the relevant portion is $t > 3$.

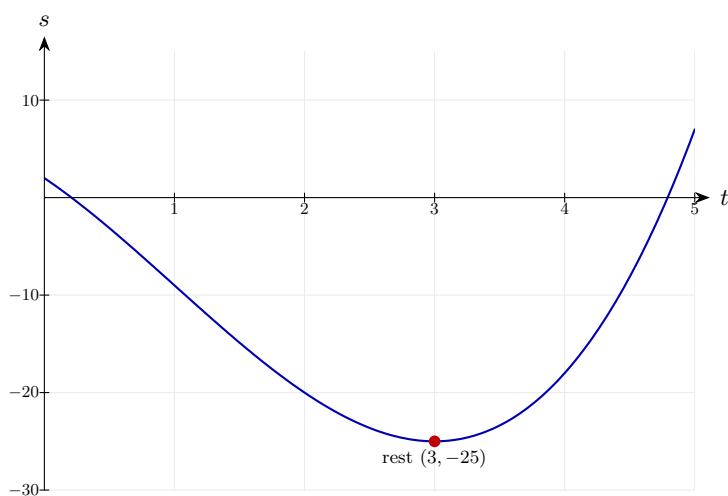
Step 4: (b) Acceleration.

$$a = \frac{dv}{dt} = 6t - 6; \quad a(4) = 24 - 6 = 18 \text{ ms}^{-2}$$

Differentiate v then substitute $t = 4$.

Final answer: (a) $v = 3(t - 3)(t + 1)$; particle moves in positive direction for $t > 3$
(b) $a(4) = 18 \text{ ms}^{-2}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 31 (d5-031)

Difficulty: Foundation / Marks: 8

A drinks can manufacturer designs a cylindrical can of volume 330 cm^3 . The radius is r cm and the height is h cm. (a) Show that the total surface area (including both circular ends) is $A = 2\pi r^2 + \frac{660}{r}$. (b) Find the value of r that minimises the surface area, giving your answer to 3 significant figures. (c) Find the minimum surface area to 3 significant figures.

Worked solution.

Step 1: (a) Volume: $\pi r^2 h = 330$.

$$h = \frac{330}{\pi r^2}$$

Express h in terms of r .

Step 2: Surface area.

$$A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \cdot \frac{330}{\pi r^2} = 2\pi r^2 + \frac{660}{r}$$

Substitute h and simplify: $\frac{2\pi r \times 330}{\pi r^2} = \frac{660}{r}$.

Step 3: (b) Differentiate.

$$\frac{dA}{dr} = 4\pi r - \frac{660}{r^2}$$

Power rule applied to each term.

Step 4: Set to zero.

$$4\pi r = \frac{660}{r^2} \implies r^3 = \frac{660}{4\pi} = \frac{165}{\pi} \implies r = \sqrt[3]{\frac{165}{\pi}} \approx 3.74 \text{ cm}$$

Multiply by r^2 and solve for r .

Step 5: Confirm minimum.

$$\frac{d^2 A}{dr^2} = 4\pi + \frac{1320}{r^3} > 0 \Rightarrow \text{minimum}$$

Both terms are positive.

Step 6: (c) Minimum surface area.

$$A \approx 2\pi(3.74)^2 + \frac{660}{3.74} \approx 87.9 + 176.5 \approx 264 \text{ cm}^2$$

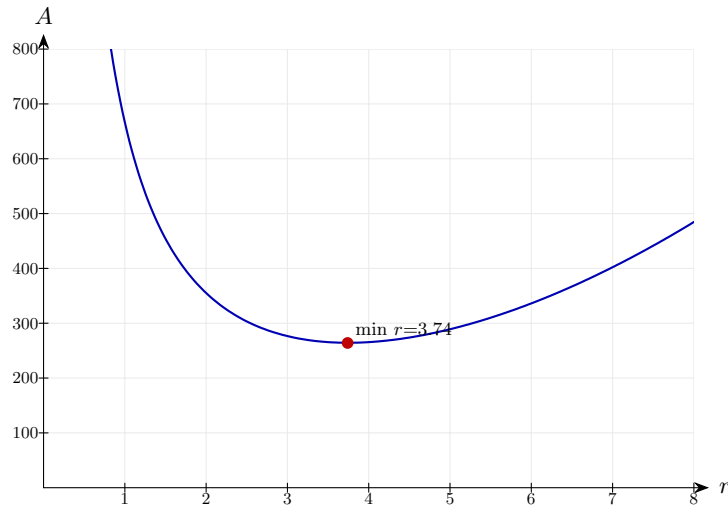
Substitute $r \approx 3.74$ into A .

Final answer: (a) Shown.

(b) $r \approx 3.74$ cm

(c) Minimum surface area $\approx 264 \text{ cm}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 32 (d5-032)

Difficulty: Foundation / Marks: 7

A particle moves along a straight line so that its displacement s metres at time t seconds is $s = 2t^3 - 5t^2 + 4t$ for $t \geq 0$. (a) Find the velocity function and determine when the particle first comes to rest. (b) Find the acceleration at that instant. (c) Determine whether the particle is speeding up or slowing down just before it comes to rest.

Worked solution.

Step 1: (a) Velocity.

$$v = 6t^2 - 10t + 4$$

Differentiate s .

Step 2: Set $v = 0$.

$$6t^2 - 10t + 4 = 0 \implies 3t^2 - 5t + 2 = 0 \implies (3t - 2)(t - 1) = 0$$

Divide by 2 and factorise.

Step 3: First time at rest.

$$t = \frac{2}{3} \text{ s (earlier root)}$$

$t = \frac{2}{3} < 1$, so the particle first comes to rest at $t = \frac{2}{3}$ s.

Step 4: (b) Acceleration.

$$a = 12t - 10; \quad a\left(\frac{2}{3}\right) = 8 - 10 = -2 \text{ ms}^{-2}$$

Differentiate v and substitute $t = \frac{2}{3}$.

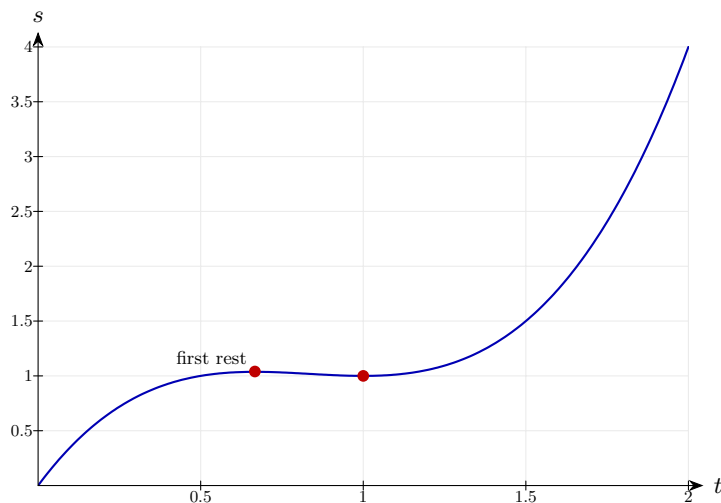
Step 5: (c) Just before $t = \frac{2}{3}$, check the sign of v .

$$v(0.5) = 6(0.25) - 10(0.5) + 4 = 1.5 - 5 + 4 = 0.5 > 0$$

The particle is moving in the positive direction. The acceleration is -2 ms^{-2} (opposing the motion), so the particle is slowing down.

Final answer: (a) $v = 6t^2 - 10t + 4$; first at rest at $t = \frac{2}{3}$ s
 (b) $a = -2 \text{ ms}^{-2}$
 (c) Slowing down (velocity positive, acceleration negative)

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 33 (d5-033)

Difficulty: Foundation / Marks: 8

An architect designs a Norman window consisting of a rectangle surmounted by a semi-circle. The perimeter of the window (including the base) is 6 m and the width of the rectangle is $2r$ m. (a) Show that the height of the rectangular part is $h = 3 - r - \frac{\pi r}{2}$. (b)

Show that the total area is $A = 6r - 2r^2 - \frac{\pi r^2}{2}$. (c) Find the value of r that maximises the area, giving your answer in exact form.

Worked solution.

Step 1: (a) Perimeter: base + two sides + semicircle arc.

$$2r + 2h + \pi r = 6 \implies 2h = 6 - 2r - \pi r \implies h = 3 - r - \frac{\pi r}{2}$$

The semicircle arc has length πr (half the circumference of a circle of radius r).

Step 2: (b) Total area = rectangle + semicircle.

$$A = 2rh + \frac{\pi r^2}{2} = 2r \left(3 - r - \frac{\pi r}{2} \right) + \frac{\pi r^2}{2}$$

Substitute h into the area expression.

Step 3: Expand.

$$A = 6r - 2r^2 - \pi r^2 + \frac{\pi r^2}{2} = 6r - 2r^2 - \frac{\pi r^2}{2}$$

$$-\pi r^2 + \frac{\pi r^2}{2} = -\frac{\pi r^2}{2}.$$

Step 4: (c) Differentiate.

$$\frac{dA}{dr} = 6 - 4r - \pi r = 6 - r(4 + \pi)$$

Power rule.

Step 5: Set to zero.

$$r(4 + \pi) = 6 \implies r = \frac{6}{4 + \pi}$$

This is the exact value.

Step 6: Confirm maximum.

$$\frac{d^2A}{dr^2} = -(4 + \pi) < 0 \Rightarrow \text{maximum}$$

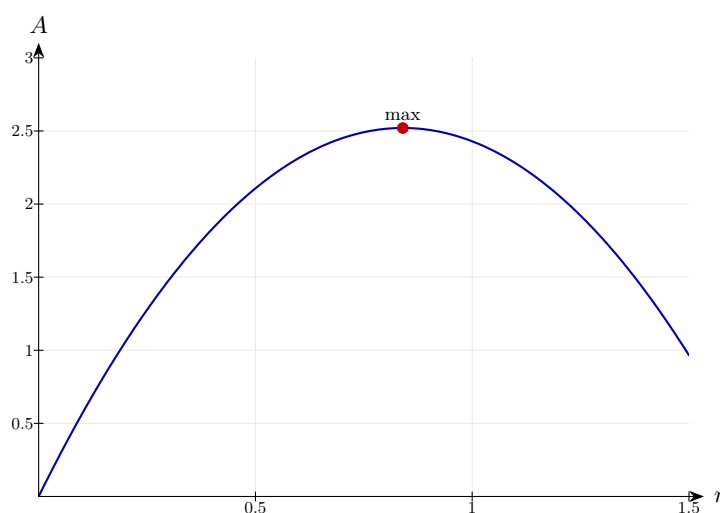
Negative second derivative.

Final answer: (a) Shown.

(b) Shown.

(c) $r = \frac{6}{4 + \pi}$ m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 34 (d5-034)

Difficulty: Foundation / Marks: 8

A particle P moves along the x -axis. Its displacement from the origin O at time t seconds is $x = t^3 - 12t + 4$ metres for $t \geq 0$. (a) Find $\frac{dx}{dt}$ and $\frac{d^2x}{dt^2}$. (b) Find the times when P is at rest. (c) Find the position of P at each of those times. (d) Determine the direction P is travelling just after $t = 2$.

Worked solution.

Step 1: (a) Velocity.

$$\frac{dx}{dt} = 3t^2 - 12$$

Power rule.

Step 2: Acceleration.

$$\frac{d^2x}{dt^2} = 6t$$

Differentiate the velocity.

Step 3: (b) Set velocity to zero.

$$3t^2 - 12 = 0 \implies t^2 = 4 \implies t = 2 \text{ s (reject } t = -2)$$

Only $t = 2$ is valid for $t \geq 0$.

Step 4: (c) Position at $t = 2$.

$$x(2) = 8 - 24 + 4 = -12 \text{ m}$$

Particle is 12 m on the negative side of O .

Step 5: (d) Just after $t = 2$, check the sign of v at, say, $t = 2.1$.

$$v(2.1) = 3(4.41) - 12 = 13.23 - 12 = 1.23 > 0$$

Positive velocity means the particle moves in the positive direction (towards O and beyond) just after $t = 2$.

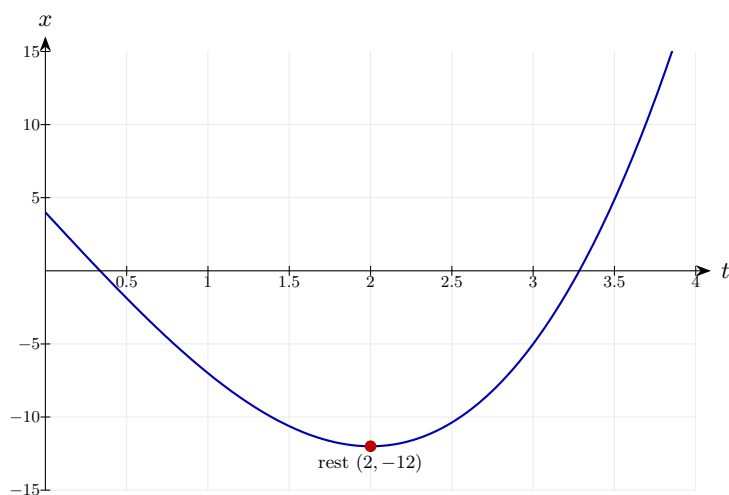
Final answer: (a) $\frac{dx}{dt} = 3t^2 - 12$, $\frac{d^2x}{dt^2} = 6t$

(b) $t = 2 \text{ s}$

(c) $x = -12 \text{ m}$

(d) Positive direction

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 35 (d5-035)

Difficulty: Foundation / Marks: 9

A company makes storage crates that are open-topped rectangular boxes. Each crate has a square base of side x m, height h m, and a volume of 4 m^3 . (a) Show that the surface

area (base and four sides) is $A = x^2 + \frac{16}{x}$. (b) Use calculus to find the value of x that minimises the surface area. (c) Find the minimum surface area and the corresponding height. (d) Show that your answer is a minimum.

Worked solution.

Step 1: (a) Volume: $x^2h = 4$.

$$h = \frac{4}{x^2}$$

Express h in terms of x .

Step 2: Surface area: base + 4 sides.

$$A = x^2 + 4xh = x^2 + 4x \cdot \frac{4}{x^2} = x^2 + \frac{16}{x}$$

Each side has area xh ; there are 4 sides.

Step 3: (b) Differentiate.

$$\frac{dA}{dx} = 2x - \frac{16}{x^2}$$

$$\frac{d}{dx} \left(\frac{16}{x} \right) = -\frac{16}{x^2}.$$

Step 4: Set to zero.

$$2x = \frac{16}{x^2} \implies 2x^3 = 16 \implies x^3 = 8 \implies x = 2 \text{ m}$$

Solve the equation.

Step 5: (c) Minimum surface area.

$$A = 4 + \frac{16}{2} = 4 + 8 = 12 \text{ m}^2$$

Substitute $x = 2$.

Step 6: Corresponding height.

$$h = \frac{4}{4} = 1 \text{ m}$$

Substitute $x = 2$ into $h = \frac{4}{x^2}$.

Step 7: (d) Second derivative test.

$$\frac{d^2A}{dx^2} = 2 + \frac{32}{x^3}; \quad x = 2 : 2 + 4 = 6 > 0 \Rightarrow \text{minimum} \checkmark$$

Positive second derivative at $x = 2$ confirms a minimum.

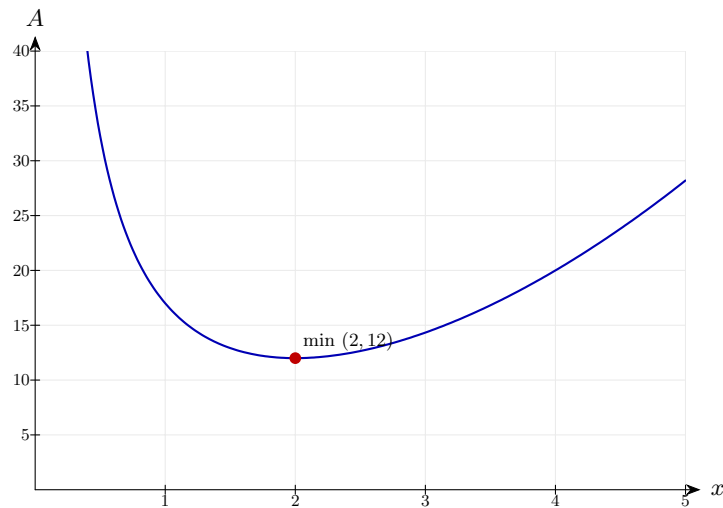
Final answer: (a) Shown.

(b) $x = 2 \text{ m}$

(c) Minimum area = 12 m^2 ; height = 1 m

(d) $\frac{d^2A}{dx^2} = 6 > 0$ at $x = 2 \checkmark$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 36 (d5-036)

Difficulty: Foundation / Marks: 4

A particle moves on a straight line with displacement $s = 2t^3 - 9t^2 + 12t$ metres after t seconds. (a) Find the velocity v . (b) Find the values of t at which the particle is instantaneously at rest.

Worked solution.

Step 1: (a) Differentiate.

$$v = \frac{ds}{dt} = 6t^2 - 18t + 12$$

Velocity is the derivative of displacement. Apply the power rule to each term of s .

Step 2: (b) Set $v = 0$.

$$6t^2 - 18t + 12 = 0 \implies t^2 - 3t + 2 = 0$$

The particle is instantaneously at rest when its velocity is zero. Divide through by 6 to simplify the quadratic.

Step 3: Factorise.

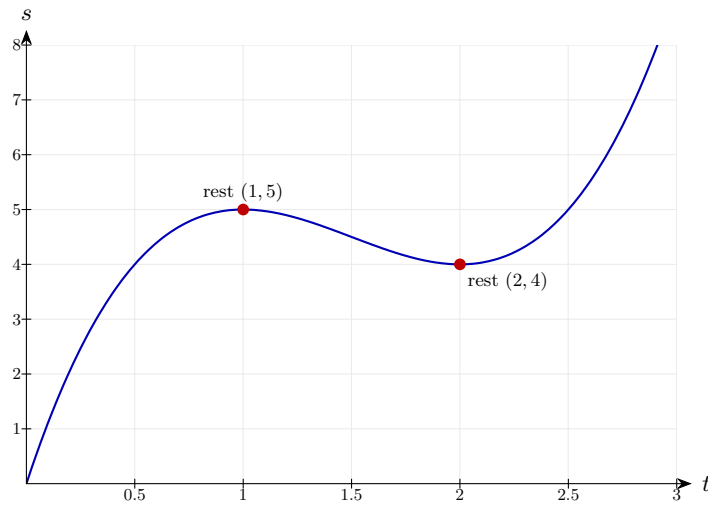
$$(t - 1)(t - 2) = 0 \implies t = 1 \text{ s or } t = 2 \text{ s}$$

Two distinct stopping moments — the particle slows, stops at $t = 1$, reverses, then stops again at $t = 2$.

Final answer: (a) $v = 6t^2 - 18t + 12$

(b) $t = 1 \text{ s}$ and $t = 2 \text{ s}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 37 (d5-037)

Difficulty: Foundation / Marks: 2

A particle has velocity $v = 6t^2 - 4t + 1 \text{ ms}^{-1}$ at time t seconds. Find the acceleration when $t = 3$.

Worked solution.

Step 1: Differentiate to get acceleration.

$$a = \frac{dv}{dt} = 12t - 4$$

Acceleration is the rate of change of velocity, so we apply the power rule to $v = 6t^2 - 4t + 1$ term by term.

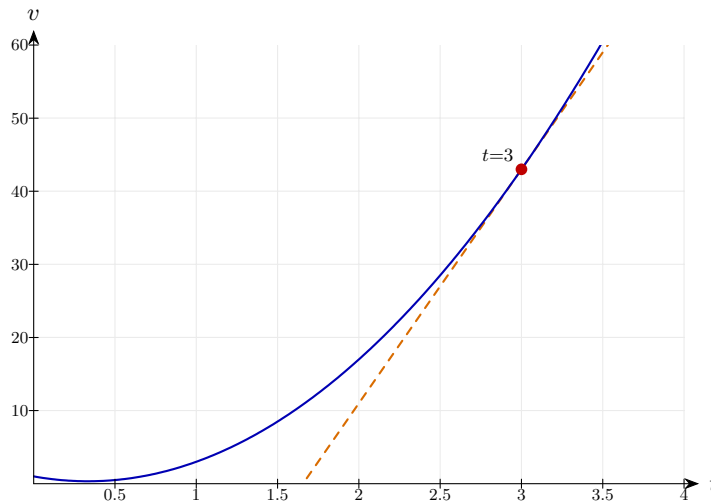
Step 2: Substitute $t = 3$.

$$a = 12(3) - 4 = 32 \text{ ms}^{-2}$$

The units ms^{-2} follow because the velocity is in ms^{-1} and we differentiated with respect to time in seconds.

Final answer: $a = 32 \text{ ms}^{-2}$

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 38 (d5-038)

Difficulty: Standard / Marks: 6

An open box is made by cutting equal squares of side x cm from the corners of a square sheet of card of side 12 cm and folding up the sides. The volume is $V = x(12 - 2x)^2$ cm³.

(a) Find $\frac{dV}{dx}$. (b) Find the value of x that maximises V and the maximum volume.

Worked solution.

Step 1: Expand.

$$V = x(144 - 48x + 4x^2) = 144x - 48x^2 + 4x^3$$

Expand $(12 - 2x)^2$ first, then multiply by x . Expanding into a polynomial makes term-by-term differentiation straightforward.

Step 2: (a) Differentiate.

$$\frac{dV}{dx} = 144 - 96x + 12x^2$$

Apply the power rule to each term.

Step 3: (b) Set to zero.

$$12x^2 - 96x + 144 = 0 \implies x^2 - 8x + 12 = 0$$

Setting $\frac{dV}{dx} = 0$ locates the stationary points of V , where the volume is neither increasing nor decreasing. Divide through by 12 to simplify.

Step 4: Factorise.

$$(x - 2)(x - 6) = 0 \implies x = 2 \text{ (since } x = 6 \text{ gives } V = 0)$$

The root $x = 6$ makes the base side $12 - 2x = 0$, giving a degenerate "box" of zero volume — reject it as it cannot be a maximum.

Step 5: Maximum volume.

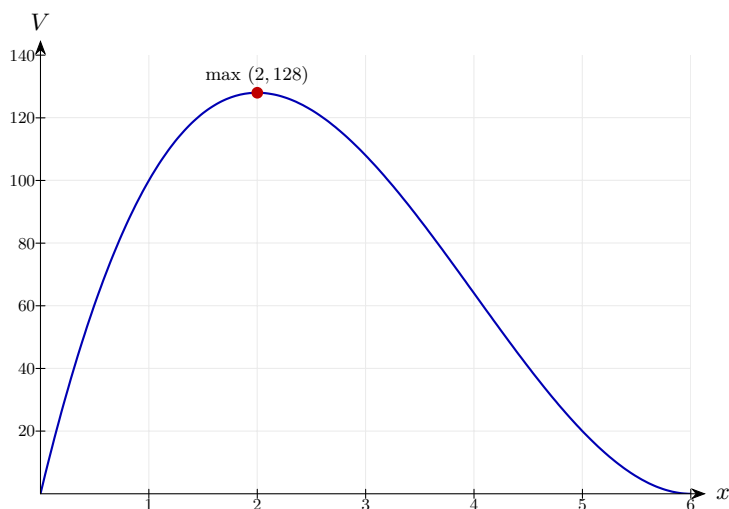
$$V = 2(12 - 4)^2 = 2 \cdot 64 = 128 \text{ cm}^3$$

Substitute $x = 2$ back into the original volume expression. (You could also check $\frac{d^2V}{dx^2} = -96 + 24x$, giving $-48 < 0$ at $x = 2$, confirming a maximum.)

Final answer: (a) $144 - 96x + 12x^2$

(b) $x = 2$ cm; $V_{\max} = 128$ cm³

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 39 (d5-039)

Difficulty: Foundation / Marks: 4

A small business has profit $P = -2x^2 + 80x - 200$ pounds when x units are sold per day. Find the value of x that maximises profit and the maximum profit.

Worked solution.

Step 1: Differentiate.

$$\frac{dP}{dx} = -4x + 80$$

This is the marginal profit — the extra profit per additional unit sold.

Step 2: Set to zero.

$$-4x + 80 = 0 \implies x = 20$$

At the maximum, marginal profit is zero — one more unit adds nothing. The negative coefficient of x^2 means P is a downward parabola, so this stationary point is guaranteed to be a maximum.

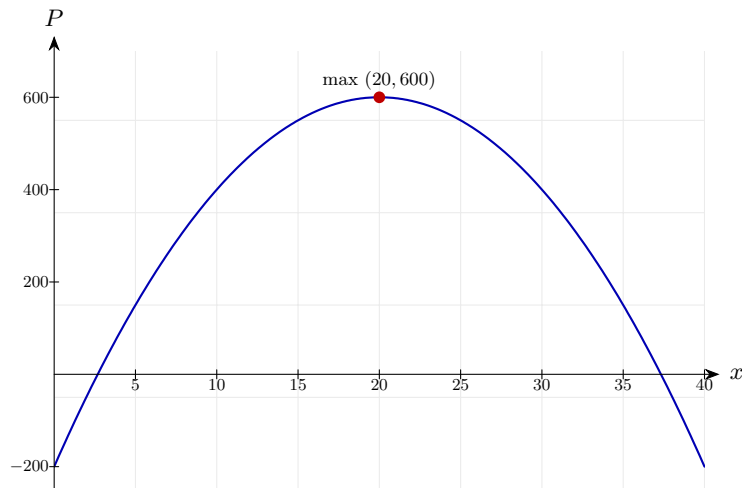
Step 3: Maximum profit.

$$P = -2(400) + 80(20) - 200 = -800 + 1600 - 200 = 600$$

Substitute $x = 20$ back into the original profit function.

Final answer: $x = 20$; maximum profit = £600

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 40 (d5-040)

Difficulty: Foundation | Marks: 5

A rectangle has perimeter 40 m. Find the dimensions that give the maximum area, and state the maximum area.

Worked solution.

Step 1: Let sides be x and y .

$$2x + 2y = 40 \implies y = 20 - x$$

Use the perimeter constraint to express y in terms of x — this reduces the area from two variables to one.

Step 2: Area as a function of x .

$$A = x(20 - x) = 20x - x^2$$

Substitute $y = 20 - x$ into $A = xy$. Now A depends on a single variable, ready for calculus.

Step 3: Differentiate.

$$\frac{dA}{dx} = 20 - 2x = 0 \implies x = 10$$

Setting the derivative to zero finds the stationary point of A . Since $A = 20x - x^2$ is a downward parabola, this stationary point is the maximum.

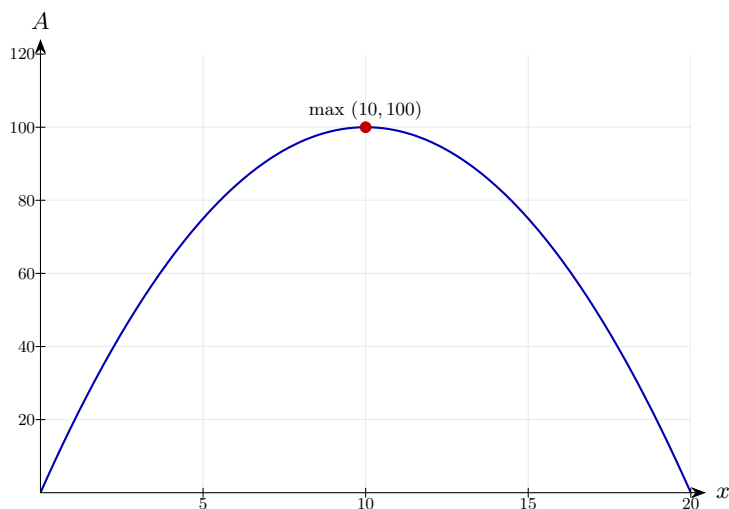
Step 4: Maximum area.

$$A = 10 \cdot 10 = 100 \text{ m}^2$$

A square of side 10 m — for a fixed perimeter, the square always gives the greatest rectangular area.

Final answer: 10 m $\times$ 10 m; area = 100 m²

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 41 (d5-041)

Difficulty: Standard / Marks: 7

A closed cylindrical can has volume $16\pi \text{ cm}^3$. Find the radius and height that minimise the total surface area, and state the minimum surface area.

Worked solution.

Step 1: Volume constraint.

$$\pi r^2 h = 16\pi \implies h = \frac{16}{r^2}$$

The fixed-volume condition links h to r . Use it to eliminate h from the surface area so we have a function of r alone.

Step 2: Total surface area.

$$A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \frac{32\pi}{r}$$

Two circular ends contribute $2\pi r^2$; the curved side contributes $2\pi r h$. Substituting $h = 16/r^2$ gives a single-variable expression.

Step 3: Differentiate.

$$\frac{dA}{dr} = 4\pi r - \frac{32\pi}{r^2}$$

Rewriting $\frac{32\pi}{r}$ as $32\pi r^{-1}$ gives derivative $-32\pi r^{-2}$. Power rule.

Step 4: Set to zero.

$$4\pi r = \frac{32\pi}{r^2} \implies r^3 = 8 \implies r = 2$$

At the optimal radius the marginal surface-area change is zero. Multiply both sides by r^2 and divide by 4π to isolate r^3 .

Step 5: Find h and A .

$$h = \frac{16}{4} = 4; \quad A = 8\pi + 16\pi = 24\pi \text{ cm}^2$$

Substitute $r = 2$ back into the constraint to find h , then into A . Notice $h = 2r$: for a closed cylinder of fixed volume, minimum surface area occurs when height equals diameter.

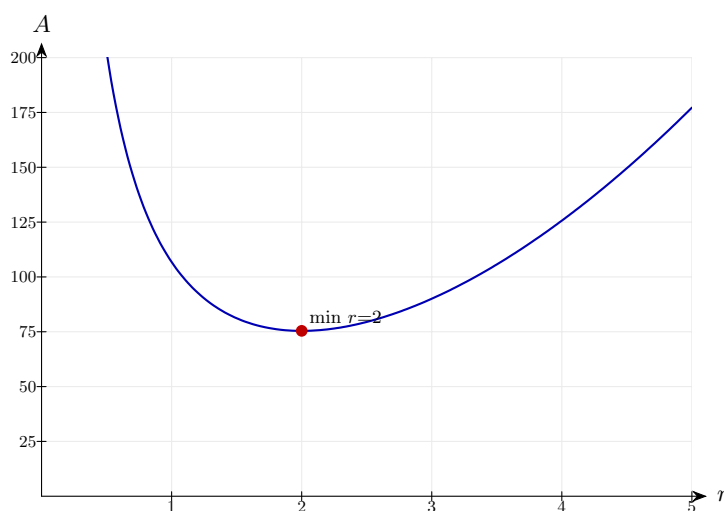
Step 6: Confirm minimum.

$$\frac{d^2A}{dr^2} = 4\pi + \frac{64\pi}{r^3} > 0 \Rightarrow \text{min} \checkmark$$

Both terms are positive for $r > 0$, so the second derivative is positive everywhere — guaranteeing the stationary point is a minimum.

Final answer: $r = 2$ cm, $h = 4$ cm, $A_{\min} = 24\pi$ cm²

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 42 (d5-042)

Difficulty: Foundation / Marks: 4

A population of insects is modelled by $P = 50t - t^2$ (in thousands) for $0 \leq t \leq 50$ days. Find the day on which the population is greatest and the maximum population.

Worked solution.

Step 1: Differentiate.

$$\frac{dP}{dt} = 50 - 2t$$

$\frac{dP}{dt}$ gives the daily rate of change of the population.

Step 2: Set to zero.

$$50 - 2t = 0 \implies t = 25$$

The population peaks when its rate of change is zero — before this, population is growing; after, it is declining. $t = 25$ lies inside the allowed window $0 \leq t \leq 50$.

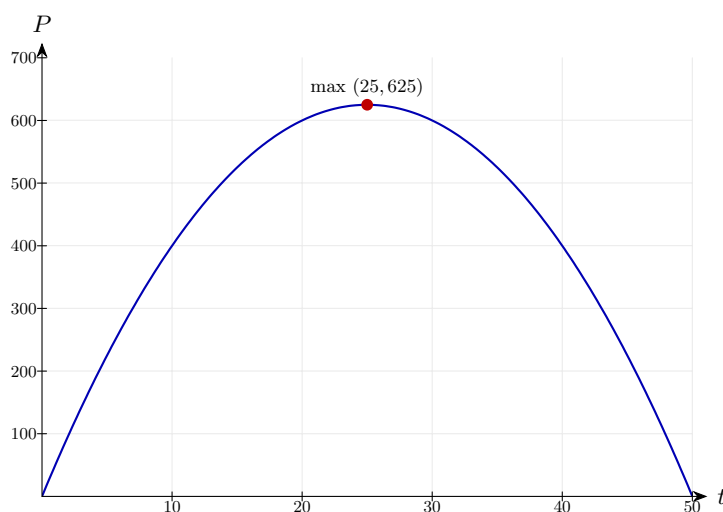
Step 3: Maximum population.

$$P = 50(25) - 25^2 = 1250 - 625 = 625 \text{ thousand}$$

Substitute $t = 25$ back into P . Since the coefficient of t^2 is negative, the curve is a downward parabola, so this stationary point is indeed the maximum.

Final answer: Day $t = 25$; maximum $P = 625,000$ insects

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 43 (d5-043)

Difficulty: Standard / Marks: 6

The cost (in pounds) of producing x units is $C(x) = x^2 + 4x + 100$. The average cost per unit is $\bar{C}(x) = \frac{C(x)}{x}$. (a) Write $\bar{C}(x)$ explicitly. (b) Find the value of x that minimises $\bar{C}$ and the minimum average cost.

Worked solution.

Step 1: (a) Divide by x .

$$\bar{C}(x) = x + 4 + \frac{100}{x}$$

Divide each term of $C(x)$ by x to get the average cost per unit. The $100/x$ term comes from the fixed cost £100 spread over x units.

Step 2: (b) Differentiate.

$$\frac{d\bar{C}}{dx} = 1 - \frac{100}{x^2}$$

Rewrite $100/x$ as $100x^{-1}$; its derivative is $-100x^{-2} = -100/x^2$.

Step 3: Set to zero.

$$1 = \frac{100}{x^2} \implies x^2 = 100 \implies x = 10$$

A stationary point of the average cost. Take the positive root only — x is a number of units.

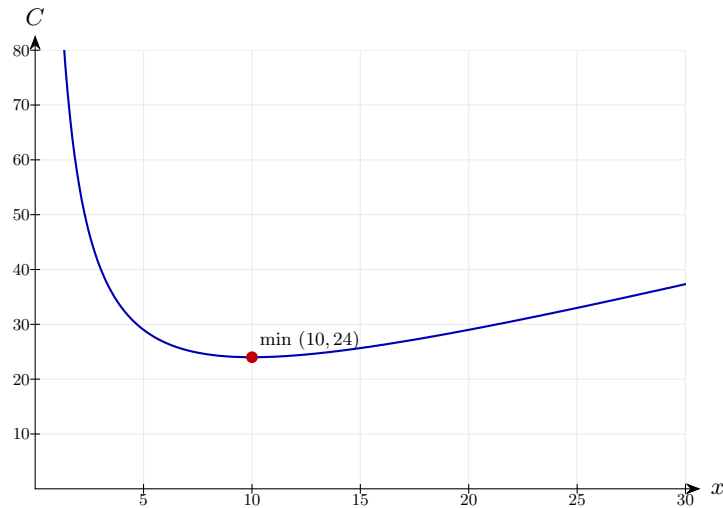
Step 4: Minimum average cost.

$$\bar{C}(10) = 10 + 4 + 10 = £24$$

Substitute $x = 10$. It is a minimum because $\frac{d^2\bar{C}}{dx^2} = 200/x^3 > 0$ for $x > 0$.

Final answer: (a) $\bar{C} = x + 4 + \frac{100}{x}$
(b) $x = 10$; minimum = £24

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 44 (d5-044)

Difficulty: Foundation / Marks: 3

The cost of producing x items is $C(x) = 0.03x^3 - x^2 + 50x + 100$ pounds. The marginal cost is $C'(x)$. Find the marginal cost when $x = 10$.

Worked solution.

Step 1: Differentiate.

$$C'(x) = 0.09x^2 - 2x + 50$$

The marginal cost $C'(x)$ is the derivative of the total cost — it approximates the cost of producing one more item beyond the x th.

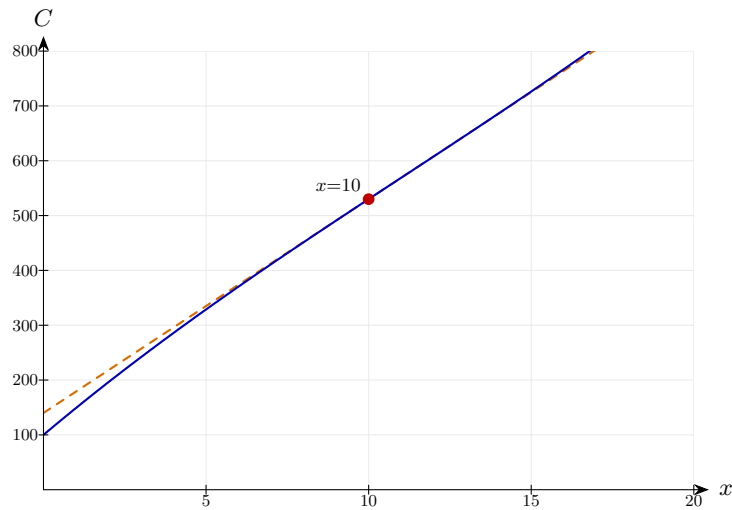
Step 2: Substitute $x = 10$.

$$C'(10) = 9 - 20 + 50 = 39$$

At a production level of 10 items, producing one more item costs about £39.

Final answer: Marginal cost = £39 per item

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 45 (d5-045)

Difficulty: Foundation / Marks: 3

Water flows into a tank so that the volume after t seconds is $V = 20t - \frac{1}{2}t^2$ litres. Find the rate at which water is flowing in when $t = 4$.

Worked solution.

Step 1: Differentiate.

$$\frac{dV}{dt} = 20 - t$$

$\frac{dV}{dt}$ gives the instantaneous flow rate — how fast the volume is changing per second.

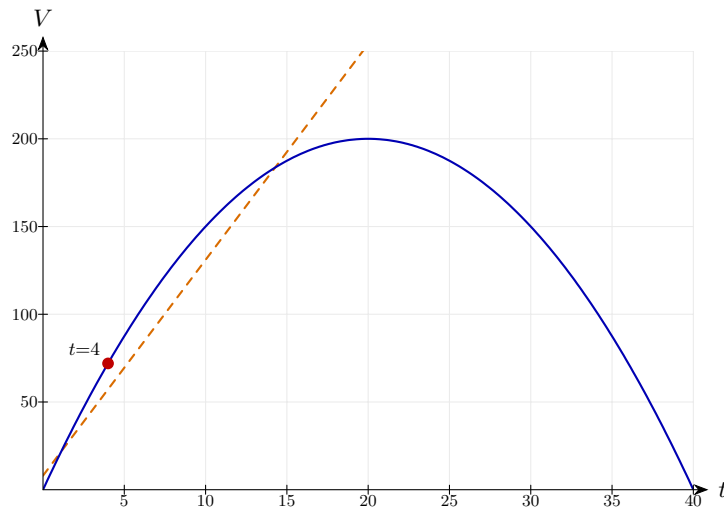
Step 2: Substitute $t = 4$.

$$\frac{dV}{dt} = 20 - 4 = 16 \text{ litres/s}$$

A positive rate means water is flowing in. The rate is decreasing with time, suggesting the inflow is slowing.

Final answer: 16 litres per second

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 46 (d5-046)

Difficulty: Standard / Marks: 5

A particle moves with displacement $s = 12t - t^3$ metres after t seconds, for $t \geq 0$. Find the maximum displacement and the time at which it occurs.

Worked solution.

Step 1: Velocity.

$$v = \frac{ds}{dt} = 12 - 3t^2$$

At maximum displacement the particle is momentarily stationary, so we will need $v = 0$.

Step 2: Set $v = 0$.

$$12 - 3t^2 = 0 \implies t^2 = 4 \implies t = 2$$

Reject $t = -2$ since $t \geq 0$. At $t = 2$ the displacement is neither increasing nor decreasing — a candidate for maximum.

Step 3: Maximum displacement.

$$s = 12(2) - 8 = 16 \text{ m}$$

Substitute $t = 2$ into the original displacement function.

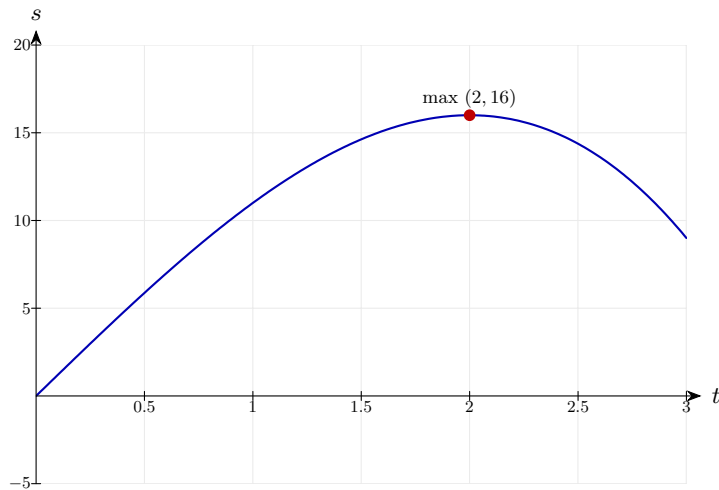
Step 4: Confirm maximum.

$$a = -6t; a(2) = -12 < 0 \Rightarrow \text{max} \checkmark$$

A negative second derivative $\left(\frac{d^2s}{dt^2}\right)$ at the stationary point confirms it is a maximum, not a minimum.

Final answer: $s_{\max} = 16 \text{ m}$ at $t = 2 \text{ s}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 47 (d5-047)

Difficulty: Foundation / Marks: 3

A firm's revenue from selling x items is $R(x) = 200x - 0.5x^2$ pounds. Find the marginal revenue when $x = 20$.

Worked solution.

Step 1: Differentiate.

$$R'(x) = 200 - x$$

Marginal revenue is the derivative of total revenue — it approximates the extra revenue from selling one more item.

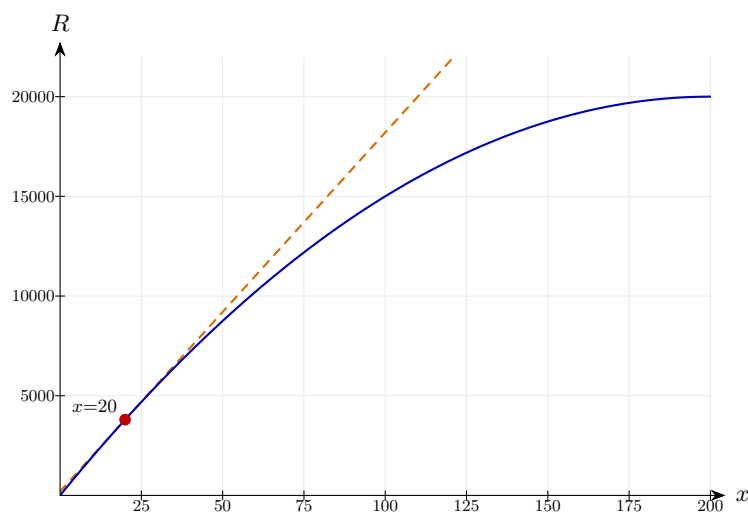
Step 2: Substitute.

$$R'(20) = 180$$

At a sales level of 20 items, the next item adds about £180 to revenue.

Final answer: Marginal revenue = £180

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 48 (d5-048)

Difficulty: Standard / Marks: 5

A stone is dropped from a cliff. Its height above the ground after t seconds is $h = 80 - 5t^2$ metres. (a) Find the time taken for the stone to reach the ground. (b) Find the speed of the stone when it hits the ground.

Worked solution.

Step 1: (a) Set $h = 0$.

$$80 - 5t^2 = 0 \implies t^2 = 16 \implies t = 4 \text{ s}$$

The stone hits the ground when its height is zero. Take the positive root since $t \geq 0$.

Step 2: (b) Velocity.

$$v = \frac{dh}{dt} = -10t$$

Velocity is the rate of change of height. The negative sign reflects that height is decreasing — the stone is moving downwards.

Step 3: Speed at $t = 4$.

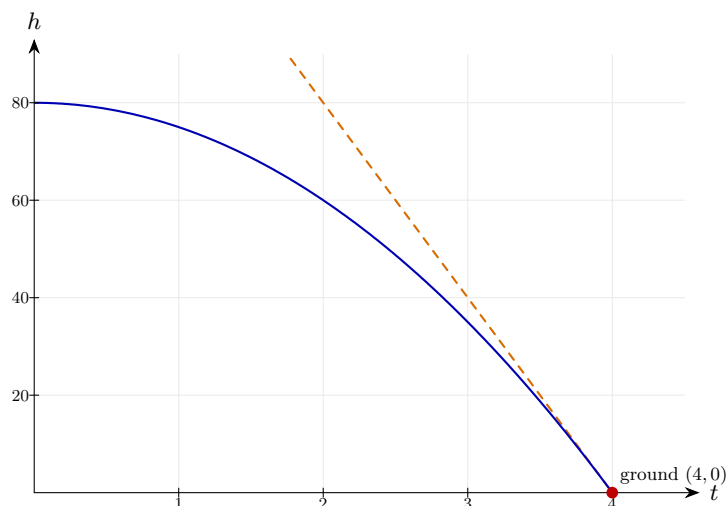
$$v = -40 \implies \text{speed} = 40 \text{ ms}^{-1}$$

Speed is the magnitude of velocity (no direction). The minus sign is dropped when reporting speed.

Final answer: (a) $t = 4 \text{ s}$

(b) 40 ms^{-1}

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 49 (d5-049)

Difficulty: Standard / Marks: 7

An open cylindrical container (no lid) has volume $27\pi \text{ cm}^3$. The radius is $r \text{ cm}$ and the height is $h \text{ cm}$. Find the radius and height that minimise the surface area.

Worked solution.

Step 1: Volume.

$$\pi r^2 h = 27\pi \implies h = \frac{27}{r^2}$$

Use the fixed-volume constraint to eliminate h , so the surface area becomes a single-variable function.

Step 2: Surface area (base + curved).

$$A = \pi r^2 + 2\pi r h = \pi r^2 + \frac{54\pi}{r}$$

There is no top (open container), so we have only one circular face plus the curved side. Substitute $h = 27/r^2$.

Step 3: Differentiate.

$$\frac{dA}{dr} = 2\pi r - \frac{54\pi}{r^2}$$

Power rule: write $\frac{54\pi}{r} = 54\pi r^{-1}$, whose derivative is $-54\pi r^{-2}$.

Step 4: Set to zero.

$$2\pi r = \frac{54\pi}{r^2} \implies r^3 = 27 \implies r = 3$$

At the minimising radius the surface area is momentarily neither growing nor shrinking. Multiply by r^2 and divide by 2π to isolate r^3 .

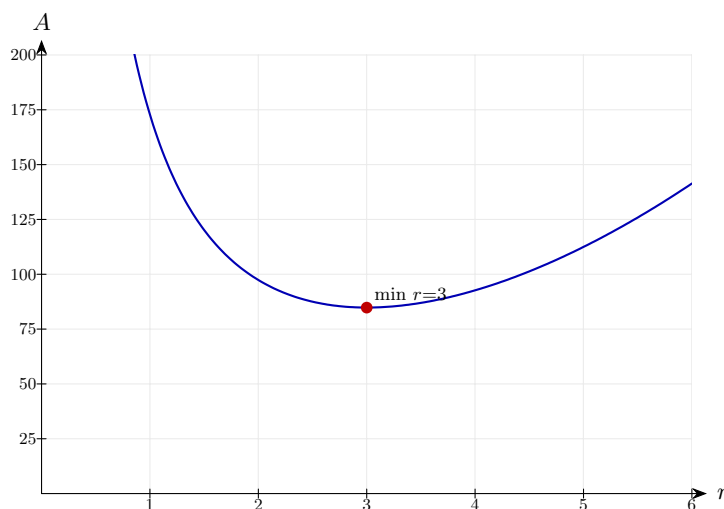
Step 5: Find h and A .

$$h = 3; \quad A = 9\pi + 18\pi = 27\pi \text{ cm}^2$$

For an open cylinder, the optimal shape has $h = r$ — height equals radius — half the height of the closed case.

Final answer: $r = 3 \text{ cm}$, $h = 3 \text{ cm}$, $A_{\min} = 27\pi \text{ cm}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 50 (d5-050)

Difficulty: Standard / Marks: 6

A farmer has 60 m of fencing and wants to enclose a rectangular area against a long straight wall, using the wall as one side. Find the dimensions and the maximum area.

Worked solution.

Step 1: Let width be w , length parallel to wall be L .

$$2w + L = 60 \implies L = 60 - 2w$$

Only three sides need fencing — two widths and one length — because the wall replaces the fourth side.

Step 2: Area.

$$A = wL = w(60 - 2w) = 60w - 2w^2$$

Substitute L into $A = wL$ to express area in one variable, ready for differentiation.

Step 3: Differentiate.

$$\frac{dA}{dw} = 60 - 4w = 0 \implies w = 15$$

A stationary point of the area. Notice the length comes out twice the width — the classic result for a three-sided enclosure.

Step 4: Find L and A .

$$L = 30; \quad A = 15 \cdot 30 = 450 \text{ m}^2$$

Substitute $w = 15$ into $L = 60 - 2w$ and then into $A = wL$.

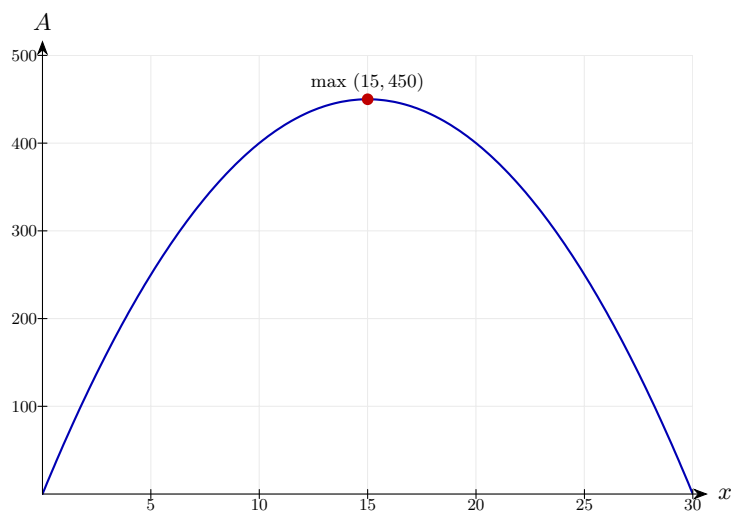
Step 5: Confirm maximum.

$$\frac{d^2A}{dw^2} = -4 < 0 \Rightarrow \text{max} \checkmark$$

Negative second derivative confirms the stationary point is a maximum, not a minimum.

Final answer: $w = 15 \text{ m}$, $L = 30 \text{ m}$, $A_{\max} = 450 \text{ m}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 51 (d5-051)

Difficulty: Standard / Marks: 6

A piece of wire of length 40 cm is cut into two pieces and each piece is bent into a square. Find the lengths so that the total area enclosed is a minimum.

Worked solution.

Step 1: Let the squares have sides x and y .

$$4x + 4y = 40 \implies y = 10 - x$$

Each square has perimeter $4 \times \text{side}$, so the two perimeters sum to 40 cm. Use this to eliminate y .

Step 2: Total area.

$$S = x^2 + (10 - x)^2 = 2x^2 - 20x + 100$$

Sum the two square areas, then expand $(10 - x)^2 = 100 - 20x + x^2$.

Step 3: Differentiate.

$$\frac{dS}{dx} = 4x - 20 = 0 \implies x = 5$$

A stationary point of total area. The positive coefficient of x^2 means S is an upward parabola, so this stationary point is a minimum (confirmed by $\frac{d^2S}{dx^2} = 4 > 0$).

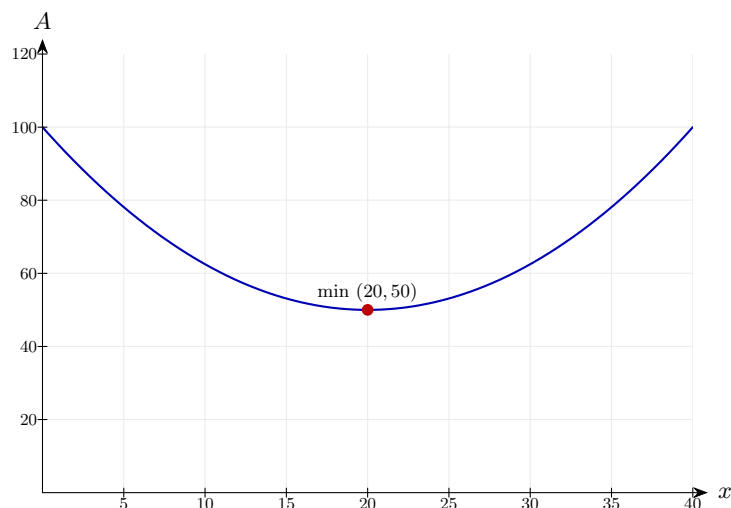
Step 4: Both pieces equal.

$$y = 5; S = 25 + 25 = 50 \text{ cm}^2$$

When the two squares are identical, surface area is minimised. Cut the wire into two equal 20 cm pieces.

Final answer: Cut into two equal 20 cm pieces; minimum area = 50 cm²

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 52 (d5-052)

Difficulty: Foundation / Marks: 4

The power output of an engine is modelled by $P = 120t - 3t^2$ watts, where t is in seconds. Find the time at which the power is greatest and the maximum power.

Worked solution.

Step 1: Differentiate.

$$\frac{dP}{dt} = 120 - 6t$$

Power output is greatest when its rate of change is zero, so we will need $\frac{dP}{dt} = 0$.

Step 2: Set to zero.

$$120 - 6t = 0 \implies t = 20$$

The negative coefficient of t^2 means P is a downward parabola, so this stationary point is the maximum.

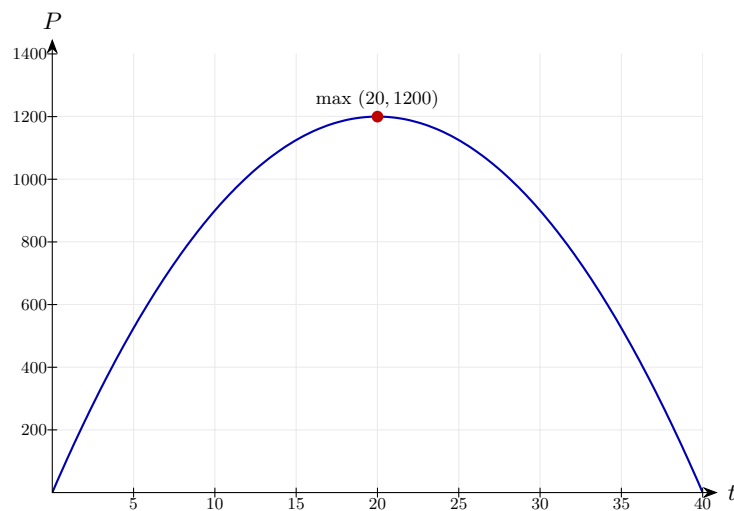
Step 3: Maximum.

$$P = 120(20) - 3(400) = 2400 - 1200 = 1200 \text{ W}$$

Substitute $t = 20$ into the original power expression.

Final answer: $t = 20$ s, $P_{\max} = 1200$ W

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 53 (d5-053)

Difficulty: Foundation / Marks: 4

A tank is being drained. The volume after t seconds is $V = 400 - 20t + \frac{t^2}{4}$ litres. Find the rate at which the volume is changing at $t = 10$ s, and interpret the sign.

Worked solution.

Step 1: Differentiate.

$$\frac{dV}{dt} = -20 + \frac{t}{2}$$

$\frac{dV}{dt}$ is the instantaneous flow rate. Apply the power rule to each term.

Step 2: Substitute.

$$\left. \frac{dV}{dt} \right|_{t=10} = -20 + 5 = -15$$

The vertical bar notation evaluates the derivative at $t = 10$.

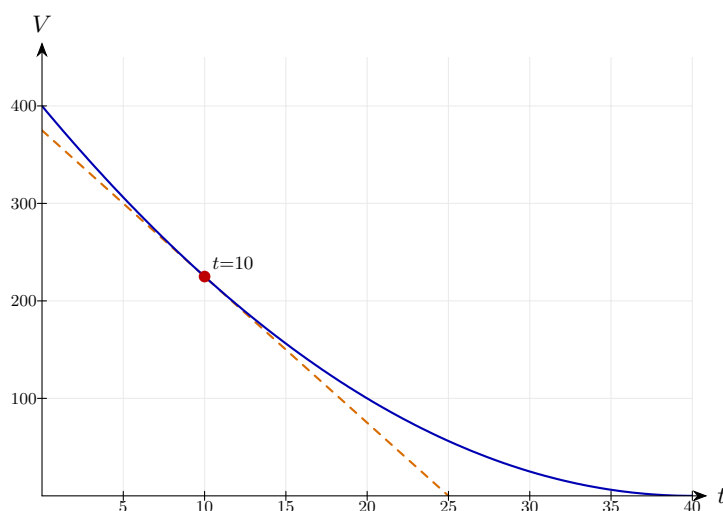
Step 3: Interpret.

Negative $\Rightarrow$ volume decreasing

A negative rate of change means the volume is shrinking — the tank is draining at 15 litres per second at this instant.

Final answer: -15 L/s; the tank is draining at 15 litres per second

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 54 (d5-054)

Difficulty: Standard / Marks: 8

A closed rectangular box has a square base of side x cm, height h cm and volume 27 cm^3 .

(a) Show that the surface area is $S = 2x^2 + \frac{108}{x}$. (b) Find the value of x that minimises S , and find the minimum surface area.

Worked solution.

Step 1: (a) Volume.

$$x^2 h = 27 \implies h = \frac{27}{x^2}$$

Fixed volume relates h to x . Use this to eliminate h from the surface area so only one variable remains.

Step 2: Surface area: 2 squares + 4 sides.

$$S = 2x^2 + 4xh = 2x^2 + \frac{108}{x} \checkmark$$

Two square ends contribute $2x^2$; the four rectangular sides each have area xh . Substitute $h = 27/x^2$ and simplify.

Step 3: (b) Differentiate.

$$\frac{dS}{dx} = 4x - \frac{108}{x^2}$$

Rewrite $\frac{108}{x}$ as $108x^{-1}$ before applying the power rule.

Step 4: Set to zero.

$$4x = \frac{108}{x^2} \implies x^3 = 27 \implies x = 3$$

The optimal base width is where the derivative vanishes. Multiply by x^2 and divide by 4 to isolate x^3 .

Step 5: Minimum surface area.

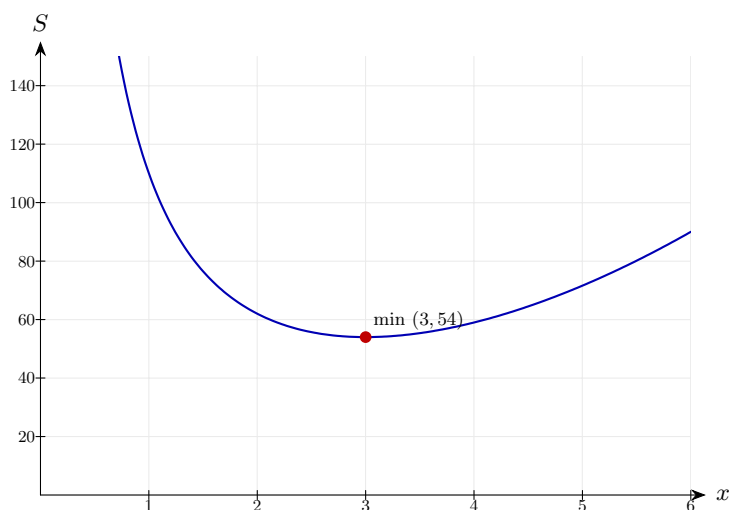
$$S = 2(9) + \frac{108}{3} = 18 + 36 = 54 \text{ cm}^2$$

Substitute $x = 3$. A check of $\frac{d^2S}{dx^2} = 4 + 216/x^3 > 0$ confirms a minimum.

Final answer: (a) Shown.

(b) $x = 3$ cm; $S_{\min} = 54 \text{ cm}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 55 (d5-055)

Difficulty: Standard / Marks: 5

A particle's displacement is $s = t^3 - 12t^2 + 36t$ metres after t seconds. Find the times at which the particle is momentarily at rest and the displacement at each.

Worked solution.

Step 1: Velocity.

$$v = 3t^2 - 24t + 36$$

Differentiate the displacement term by term. The particle is at rest when $v = 0$.

Step 2: Set $v = 0$.

$$t^2 - 8t + 12 = 0 \implies (t - 2)(t - 6) = 0$$

Divide through by 3 first to make the factorising easier.

Step 3: Times.

$$t = 2 \text{ s or } t = 6 \text{ s}$$

Two distinct moments of rest — the particle slows, stops, reverses, slows again, and stops again.

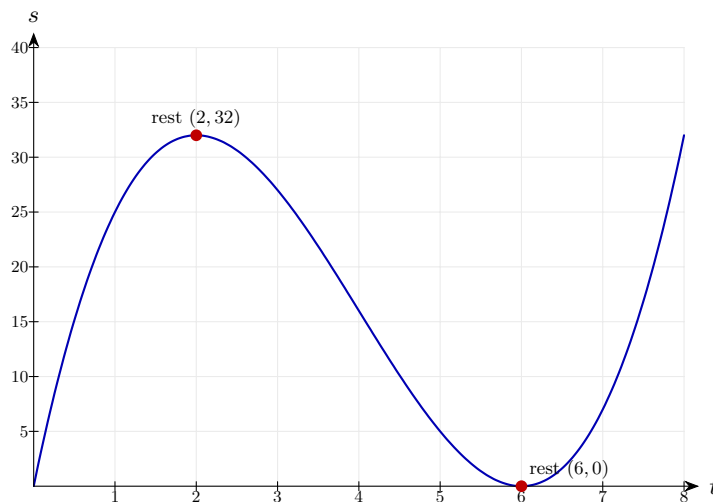
Step 4: Displacements.

$$s(2) = 8 - 48 + 72 = 32; s(6) = 216 - 432 + 216 = 0$$

Substitute each rest time back into s . The particle returns to its starting point at $t = 6$.

Final answer: $t = 2$: $s = 32$ m; $t = 6$: $s = 0$ m

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 56 (d5-056)

Difficulty: Foundation / Marks: 3

In a free-fall model, distance fallen is $s = 4.9t^2$ metres after t seconds. Find the speed of the falling object after 5 seconds.

Worked solution.

Step 1: Differentiate.

$$v = \frac{ds}{dt} = 9.8t$$

Speed is the rate of change of distance. The constant 9.8 here is the acceleration due to gravity, $g \approx 9.8 \text{ ms}^{-2}$.

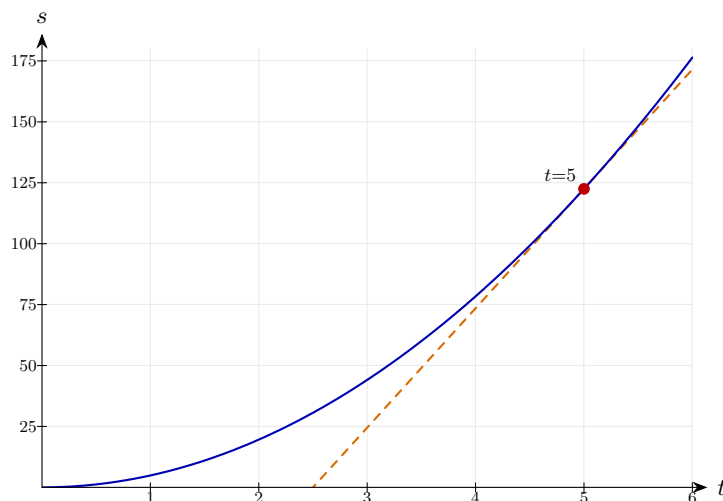
Step 2: Substitute.

$$v = 9.8 \times 5 = 49 \text{ ms}^{-1}$$

After 5 seconds of free fall the object is travelling at 49 m/s.

Final answer: 49 ms^{-1}

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 57 (d5-057)

Difficulty: Foundation / Marks: 4

A car has displacement $s = t^3 - 3t^2 - 9t + 5$ metres after t seconds, for $t \geq 0$. Find the time at which the car is momentarily at rest.

Worked solution.

Step 1: Velocity.

$$v = 3t^2 - 6t - 9 = 3(t^2 - 2t - 3)$$

Differentiate s and factor out the common 3 to simplify the quadratic.

Step 2: Factorise.

$$3(t - 3)(t + 1) = 0$$

Factor the quadratic. The car is momentarily at rest when $v = 0$.

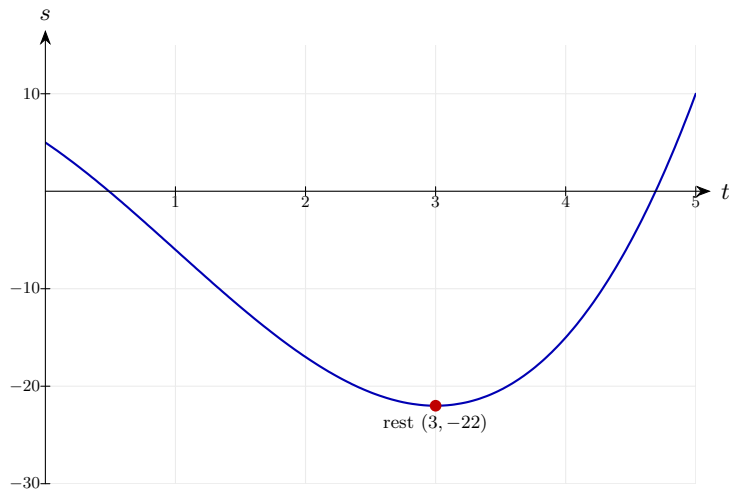
Step 3: Solve.

$$t = 3 \text{ (rejecting } t = -1 \text{ since } t \geq 0)$$

Time cannot be negative, so the only physically valid solution is $t = 3$ s.

Final answer: $t = 3$ s

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 58 (d5-058)

Difficulty: Standard / Marks: 6

A farmer has 200 m of fencing to build a rectangular field bordered on one side by a straight river (no fence on the river side). Find the maximum area she can enclose.

Worked solution.

Step 1: Let x be the width, y the length.

$$2x + y = 200 \implies y = 200 - 2x$$

The river replaces one side of the rectangle, so only three sides need fencing — two widths plus one length.

Step 2: Area.

$$A = xy = 200x - 2x^2$$

Substitute $y = 200 - 2x$ into $A = xy$ to reduce to one variable.

Step 3: Differentiate.

$$\frac{dA}{dx} = 200 - 4x = 0 \implies x = 50$$

Stationary point of the area. Negative coefficient of x^2 guarantees this is a maximum.

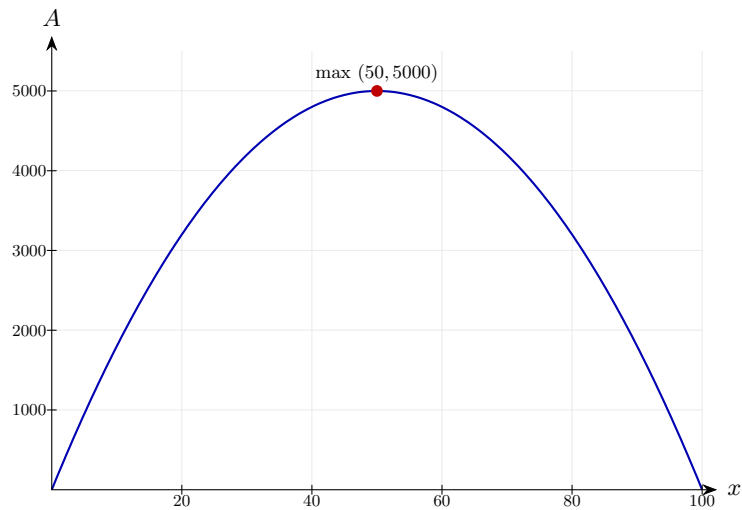
Step 4: Maximum area.

$$y = 100; A = 50 \cdot 100 = 5000 \text{ m}^2$$

Substitute back to find y and compute the area. Note $y = 2x$ — the length parallel to the river is twice the depth.

Final answer: $x = 50 \text{ m}$, $y = 100 \text{ m}$, $A_{\max} = 5000 \text{ m}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 59 (d5-059)

Difficulty: Foundation / Marks: 4

A company's profit (in pounds) on x units sold is $P(x) = 60x - x^2 - 200$. Find the number of units that maximises profit and the maximum profit.

Worked solution.

Step 1: Differentiate.

$$P'(x) = 60 - 2x$$

The marginal profit $P'(x)$ tells us how profit changes per additional unit sold.

Step 2: Set to zero.

$$60 - 2x = 0 \implies x = 30$$

At maximum profit, the marginal profit is zero — selling one more unit adds nothing. Negative coefficient of x^2 confirms this is a maximum.

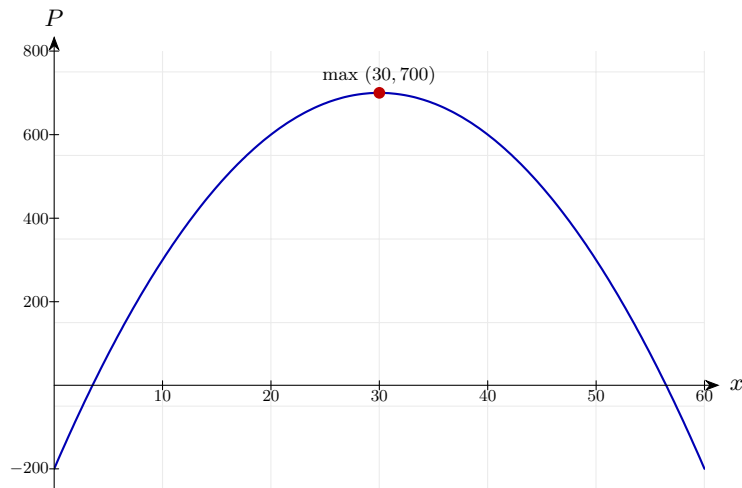
Step 3: Maximum profit.

$$P = 1800 - 900 - 200 = 700$$

Substitute $x = 30$ into $P(x) = 60x - x^2 - 200$.

Final answer: $x = 30$; maximum profit = £700

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 60 (d5-060)

Difficulty: Foundation / Marks: 3

A firm's profit is $P(x) = 20x - 0.1x^2 - 50$ pounds, where x is the number of items sold. Find the marginal profit when $x = 40$.

Worked solution.

Step 1: Differentiate.

$$P'(x) = 20 - 0.2x$$

Marginal profit is the derivative of total profit — it approximates the extra profit from the next item sold.

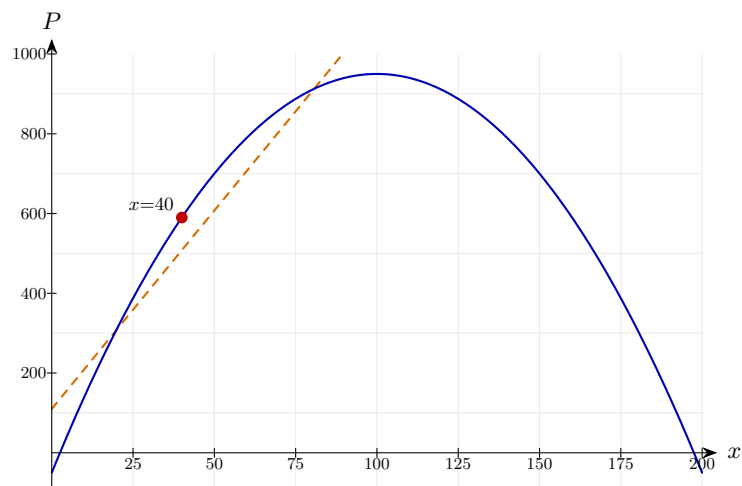
Step 2: Substitute.

$$P'(40) = 20 - 8 = 12$$

At a sales level of 40 items, the next item adds about £12 to profit. Positive marginal profit means total profit is still growing.

Final answer: Marginal profit = £12 per item

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 61 (d5-061)

Difficulty: Standard / Marks: 6

A particle's displacement is $s = t^3 - 6t^2 + 9t + 2$ metres after t seconds. Find the acceleration at each instant the particle is momentarily at rest.

Worked solution.

Step 1: Velocity.

$$v = 3t^2 - 12t + 9$$

Differentiate the displacement. The particle is at rest where $v = 0$.

Step 2: Set $v = 0$.

$$t^2 - 4t + 3 = 0 \implies (t - 1)(t - 3) = 0$$

Divide through by 3 to simplify, then factorise. Both roots are positive, so both are valid rest times.

Step 3: Acceleration.

$$a = 6t - 12$$

Differentiate v to get acceleration.

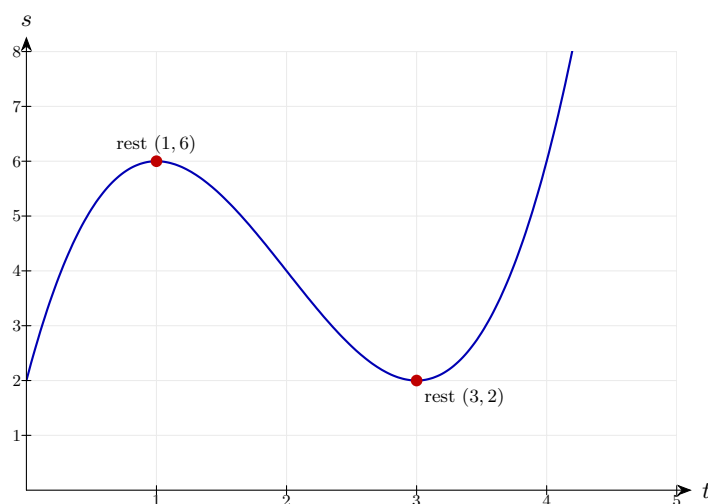
Step 4: At each rest time.

$$a(1) = -6 \text{ ms}^{-2}; a(3) = 6 \text{ ms}^{-2}$$

Negative acceleration at $t = 1$ means the particle is decelerating and about to reverse; positive at $t = 3$ means it is about to accelerate forward again.

Final answer: $a = -6 \text{ ms}^{-2}$ at $t = 1$; $a = 6 \text{ ms}^{-2}$ at $t = 3$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 62 (d5-062)

Difficulty: Standard / Marks: 7

An open cylindrical container (no lid) has volume $8\pi \text{ cm}^3$. Find the radius and height that minimise the surface area, and the minimum surface area.

Worked solution.

Step 1: Volume.

$$\pi r^2 h = 8\pi \implies h = \frac{8}{r^2}$$

The fixed-volume condition links h to r . Use it to eliminate h so the surface area depends on one variable.

Step 2: Surface area.

$$A = \pi r^2 + 2\pi r h = \pi r^2 + \frac{16\pi}{r}$$

Only one circular face (no lid) plus the curved side. Substitute $h = 8/r^2$ and simplify.

Step 3: Differentiate.

$$\frac{dA}{dr} = 2\pi r - \frac{16\pi}{r^2}$$

Write $\frac{16\pi}{r} = 16\pi r^{-1}$, then apply the power rule.

Step 4: Set to zero.

$$2\pi r^3 = 16\pi \implies r^3 = 8 \implies r = 2$$

Multiply by r^2 to clear the fraction. At the optimal radius, the surface area is momentarily not changing.

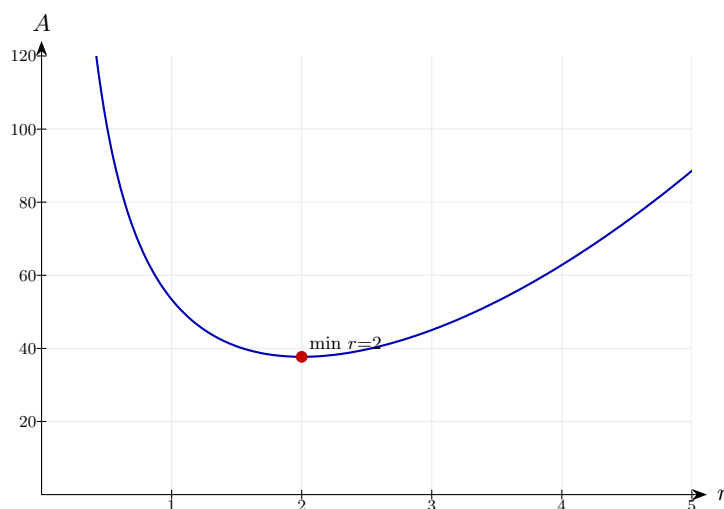
Step 5: Find h and A .

$$h = 2; A = 4\pi + 8\pi = 12\pi \text{ cm}^2$$

For an open-topped cylinder of fixed volume, the optimal shape has $h = r$.

Final answer: $r = 2 \text{ cm}$, $h = 2 \text{ cm}$, $A_{\min} = 12\pi \text{ cm}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 63 (d5-063)

Difficulty: Foundation / Marks: 4

The temperature inside an oven t minutes after switching on is modelled by $T = 20 + 30t - t^2$ °C, for $0 \leq t \leq 30$. Find the maximum temperature reached and the time at which it occurs.

Worked solution.

Step 1: Differentiate.

$$\frac{dT}{dt} = 30 - 2t$$

$\frac{dT}{dt}$ gives the rate at which the oven is heating (or cooling).

Step 2: Set to zero.

$$30 - 2t = 0 \implies t = 15$$

Temperature peaks when its rate of change is zero. $t = 15$ lies inside $0 \leq t \leq 30$, so it is a valid candidate.

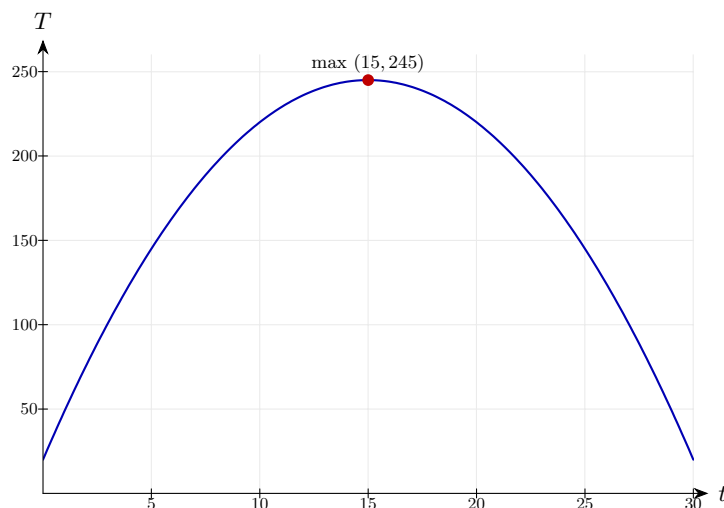
Step 3: Maximum.

$$T = 20 + 30(15) - 225 = 20 + 450 - 225 = 245$$

Substitute $t = 15$ into T . The negative coefficient of t^2 guarantees this is a maximum.

Final answer: $T_{\max} = 245^\circ\text{C}$ at $t = 15$ min

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 64 (d5-064)

Difficulty: Standard / Marks: 7

A ball is thrown upwards. Its height (in metres) above the ground after t seconds is $h = 2 + 10t - 5t^2$. (a) Find the maximum height reached. (b) Find the time at which the ball hits the ground. (c) Find the speed at which the ball hits the ground.

Worked solution.*Step 1: (a) Velocity.*

$$v = 10 - 10t = 0 \implies t = 1$$

At maximum height the ball is momentarily stationary, so $v = \frac{dh}{dt} = 0$.

Step 2: Maximum height.

$$h = 2 + 10 - 5 = 7 \text{ m}$$

Substitute $t = 1$ into h . Note this is height above the ground, not above the launch point — the ball started at $h(0) = 2 \text{ m}$.

Step 3: (b) Set $h = 0$.

$$5t^2 - 10t - 2 = 0 \implies t = \frac{10 \pm \sqrt{100 + 40}}{10}$$

The ball hits the ground when its height equals zero. Rearrange to standard form and use the quadratic formula.

Step 4: Take positive root.

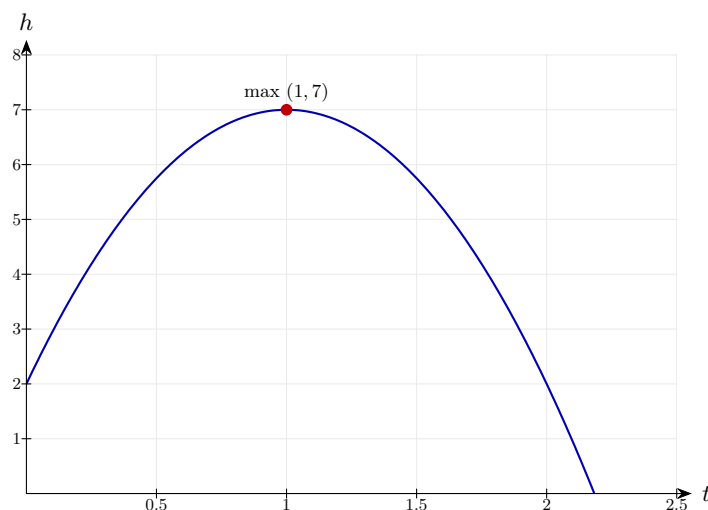
$$t = \frac{10 + \sqrt{140}}{10} \approx 2.183 \text{ s}$$

The negative root corresponds to a time before the throw and is unphysical.

Step 5: (c) Speed.

$$v = 10 - 10(2.183) \approx -11.83; |v| \approx 11.83 \text{ ms}^{-1}$$

Substitute the landing time into v . The negative sign means the ball is moving downward; speed is the magnitude.

Final answer: (a) 7 m(b) $t \approx 2.18 \text{ s}$ (c) $\approx 11.8 \text{ ms}^{-1}$ **Diagram.**

The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 65 (d5-065)

Difficulty: Standard / Marks: 4

A particle has displacement $s = t^3 - 3t$ metres after t seconds. Find the minimum velocity for $t \geq 0$.

Worked solution.

Step 1: Velocity.

$$v = 3t^2 - 3$$

Differentiate displacement to get velocity. To find the minimum velocity we treat v as the function to optimise.

Step 2: Differentiate again.

$$a = \frac{dv}{dt} = 6t$$

The derivative of v tells us whether v is increasing or decreasing. Setting $a = 0$ gives $t = 0$, which is the boundary of the domain.

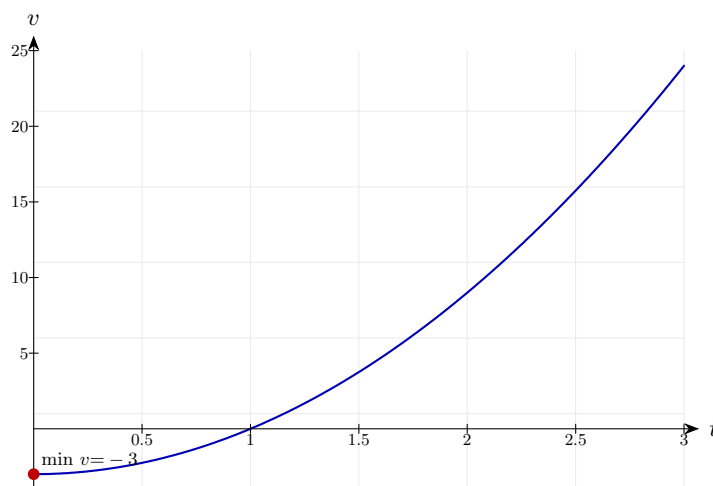
Step 3: For $t \geq 0$, $a \geq 0$, so v is minimised at $t = 0$.

$$v_{\min} = 3(0) - 3 = -3 \text{ ms}^{-1}$$

Since $a \geq 0$ on the whole domain, v is non-decreasing, so its minimum on $t \geq 0$ occurs at the left endpoint. Always check endpoints on restricted domains.

Final answer: $v_{\min} = -3 \text{ ms}^{-1}$ at $t = 0$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 66 (d5-066)

Difficulty: Standard / Marks: 6

A manufacturer's total cost (in pounds) of producing N items per day is $C = 2N^2 + 10N + 800$. The average cost per item is $\bar{C} = \frac{C}{N}$. Find the value of N that minimises the average cost and the minimum average cost.

Worked solution.*Step 1: Average cost.*

$$\bar{C} = 2N + 10 + \frac{800}{N}$$

Divide each term of C by N . The $800/N$ term represents the £800 fixed cost spread across N items.

Step 2: Differentiate.

$$\frac{d\bar{C}}{dN} = 2 - \frac{800}{N^2}$$

Rewrite $800/N = 800N^{-1}$; its derivative is $-800N^{-2}$.

Step 3: Set to zero.

$$2 = \frac{800}{N^2} \implies N^2 = 400 \implies N = 20$$

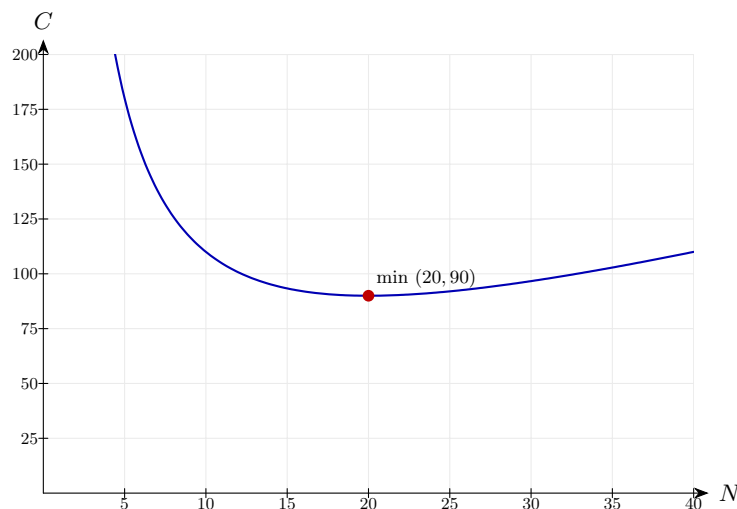
Stationary point of the average cost. Take the positive root since N is a count.

Step 4: Minimum.

$$\bar{C} = 40 + 10 + 40 = £90$$

Substitute $N = 20$. $\frac{d^2\bar{C}}{dN^2} = 1600/N^3 > 0$, confirming this is a minimum.

Final answer: $N = 20$; minimum average cost = £90

Diagram.

The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 67 (d5-067)*Difficulty: Standard / Marks: 6*

A rectangular postcard has area 100 cm^2 . Find the dimensions that minimise the perimeter and state the minimum perimeter.

Worked solution.*Step 1: Constraint.*

$$xy = 100 \implies y = \frac{100}{x}$$

Use the fixed-area constraint to express y in terms of x , so the perimeter becomes a function of one variable.

Step 2: Perimeter.

$$P = 2x + 2y = 2x + \frac{200}{x}$$

Substitute $y = 100/x$ into $P = 2x + 2y$.

Step 3: Differentiate.

$$\frac{dP}{dx} = 2 - \frac{200}{x^2}$$

Write $200/x = 200x^{-1}$, then differentiate with the power rule.

Step 4: Set to zero.

$$2 = \frac{200}{x^2} \implies x^2 = 100 \implies x = 10$$

Take the positive root since side lengths are positive.

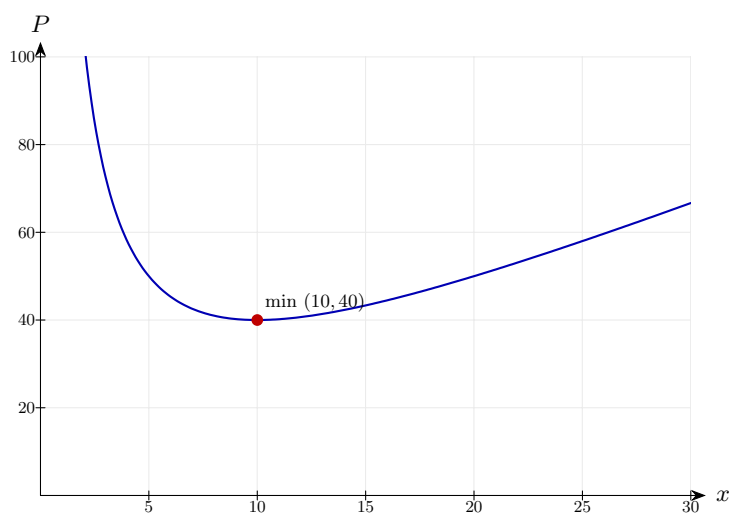
Step 5: Minimum perimeter.

$$y = 10; P = 40 \text{ cm}$$

For a fixed area, the square always gives the smallest perimeter. $\frac{d^2P}{dx^2} = 400/x^3 > 0$ confirms minimum.

Final answer: 10×10 cm; minimum perimeter = 40 cm

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 68 (d5-068)

Difficulty: Foundation / Marks: 4

A ball rolls down a slope. Its distance travelled is $s = 2t^2 + 3t$ metres after t seconds. (a) Find the velocity after t seconds. (b) Find the velocity at $t = 3$. (c) Find the acceleration.

Worked solution.

Step 1: (a) Velocity.

$$v = 4t + 3$$

Differentiate $s = 2t^2 + 3t$ using the power rule. The constant 3 represents the initial velocity at $t = 0$.

Step 2: (b) Substitute.

$$v(3) = 12 + 3 = 15 \text{ ms}^{-1}$$

Replace t with 3 in the velocity expression.

Step 3: (c) Acceleration.

$$a = 4 \text{ ms}^{-2}$$

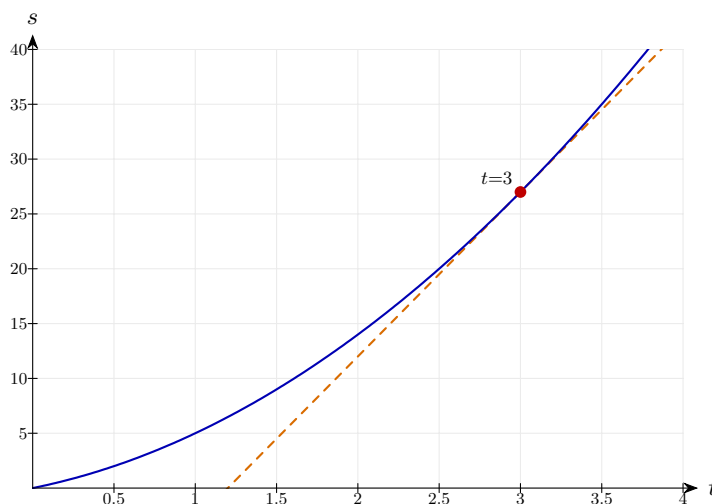
Differentiate v . Since v is linear, the acceleration is a constant — uniform acceleration.

Final answer: (a) $v = 4t + 3$

(b) 15 ms^{-1}

(c) 4 ms^{-2}

Diagram.



The model plotted against its variable, with the key point(s) marked and the tangent (dashed) showing the instantaneous rate.

Real-Life Problems 69 (d5-069)

Difficulty: Standard / Marks: 5

A particle moves with displacement $s = 2t^2 - \frac{t^3}{3}$ metres for $t \geq 0$. Find the maximum velocity.

Worked solution.

Step 1: Velocity.

$$v = 4t - t^2$$

Differentiate the displacement to get velocity. To find the maximum velocity we now treat v as the function to optimise.

Step 2: For max velocity, set $a = 0$.

$$a = 4 - 2t = 0 \implies t = 2$$

Velocity is greatest where its rate of change (the acceleration) is zero.

Step 3: Confirm max.

$$\frac{da}{dt} = -2 < 0 \Rightarrow v \text{ is max} \checkmark$$

The second derivative of v is negative, confirming v has a maximum (not a minimum) at $t = 2$.

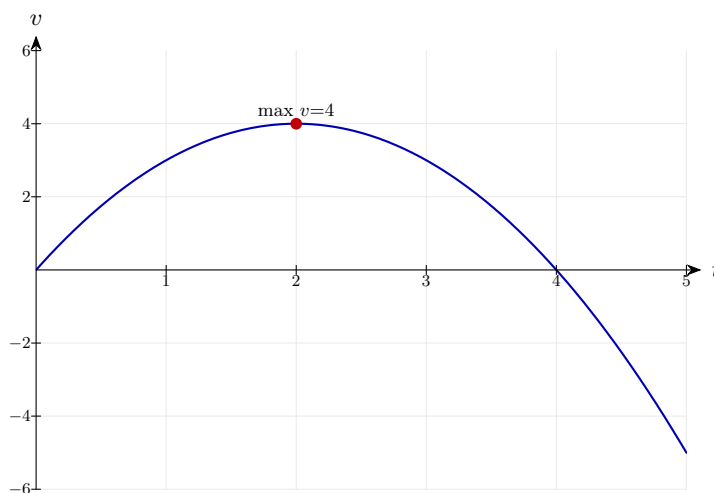
Step 4: Maximum velocity.

$$v = 8 - 4 = 4 \text{ ms}^{-1}$$

Substitute $t = 2$ into the velocity expression.

Final answer: $v_{\max} = 4 \text{ ms}^{-1}$ at $t = 2 \text{ s}$

Diagram.



The model plotted against its variable, with the key point(s) marked.

Real-Life Problems 70 (d5-070)

Difficulty: Standard / Marks: 8

An open-topped box has a rectangular base whose length is twice its width w cm, and height h cm. The volume is 36 cm^3 . (a) Show that the surface area is $S = 2w^2 + \frac{108}{w}$. (b) Find the value of w that minimises S , and find the minimum surface area.

Worked solution.

Step 1: (a) Volume.

$$2w \cdot w \cdot h = 36 \implies h = \frac{18}{w^2}$$

Length $\times$ width $\times$ height with length $= 2w$. Use the fixed-volume condition to eliminate h .

Step 2: Surface area = base + 4 sides.

$$S = 2w^2 + 2(2w)h + 2(w)h = 2w^2 + 6wh$$

Base area is $2w \times w = 2w^2$; two pairs of side faces — two have area $2wh$, two have area wh . Open-topped, so no lid.

Step 3: Substitute h .

$$S = 2w^2 + 6w \cdot \frac{18}{w^2} = 2w^2 + \frac{108}{w} \checkmark$$

Substituting h reduces S to a single-variable function of w .

Step 4: (b) Differentiate.

$$\frac{dS}{dw} = 4w - \frac{108}{w^2}$$

Rewrite $108/w$ as $108w^{-1}$ before differentiating.

Step 5: Set to zero.

$$4w^3 = 108 \implies w^3 = 27 \implies w = 3$$

At the minimising width the surface area is momentarily stationary. Multiply by w^2 to clear the fraction.

Step 6: Minimum surface area.

$$S = 18 + 36 = 54 \text{ cm}^2$$

Substitute $w = 3$ into $S = 2w^2 + 108/w$.

Step 7: Confirm minimum.

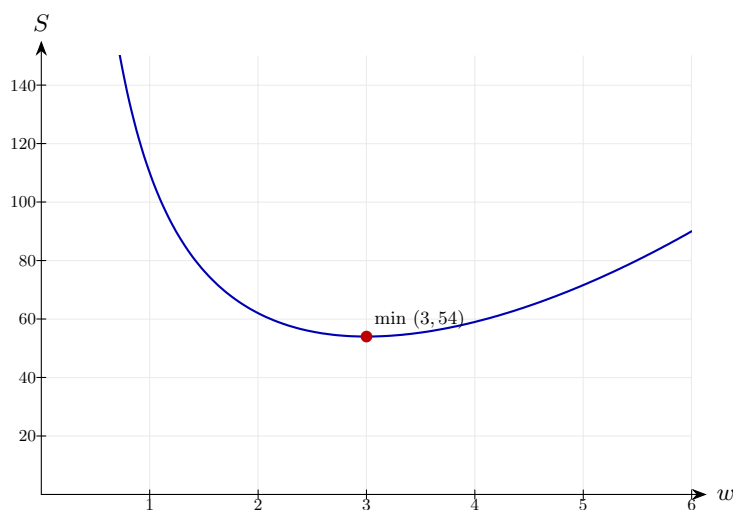
$$\frac{d^2S}{dw^2} = 4 + \frac{216}{w^3} > 0 \implies \text{min} \checkmark$$

Both terms are positive for $w > 0$, so the second derivative is positive — guaranteeing a minimum.

Final answer: (a) Shown.

(b) $w = 3 \text{ cm}$; $S_{\min} = 54 \text{ cm}^2$

Diagram.



The model plotted against its variable, with the key point(s) marked.
